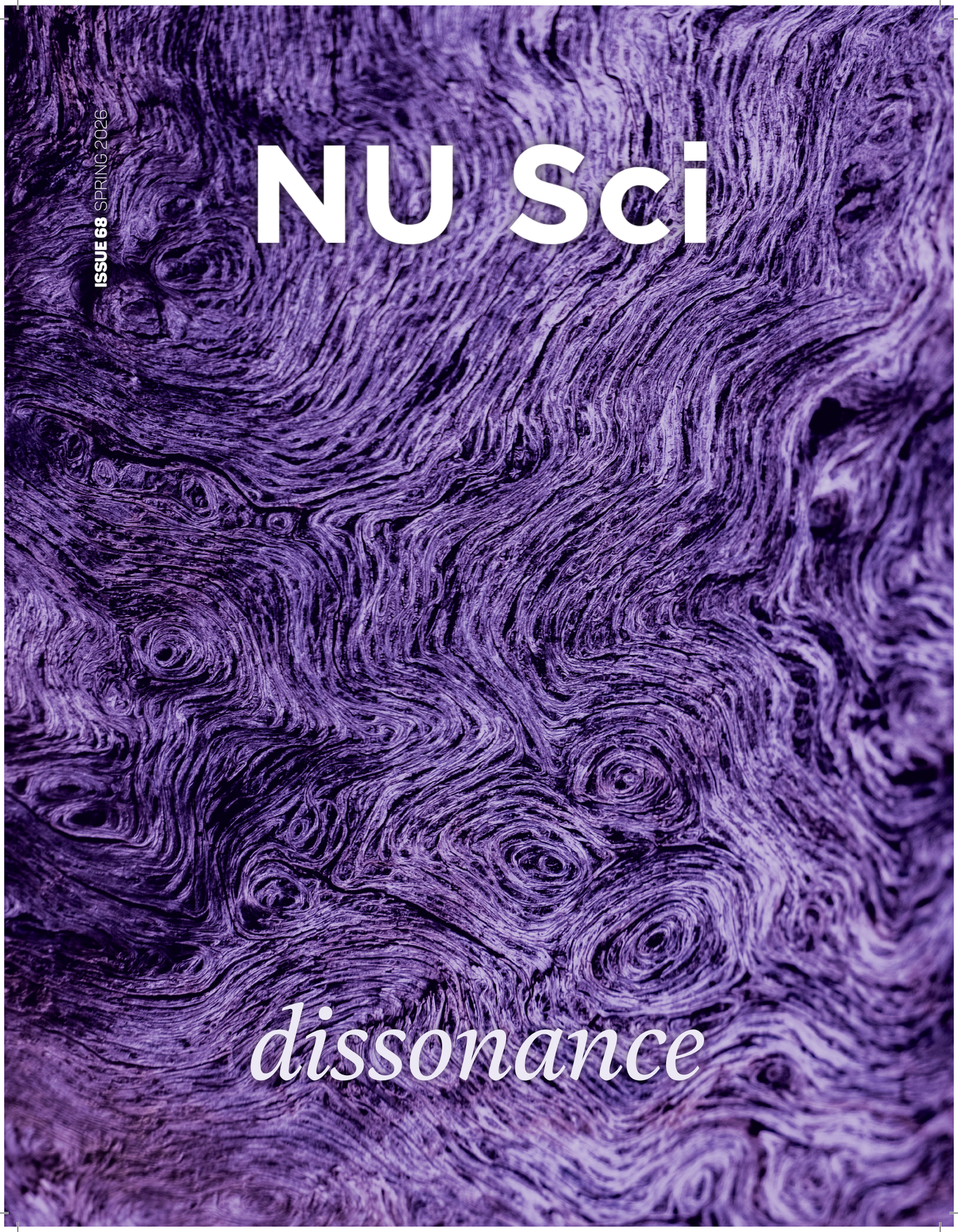
ISSUE 68 SPRING 2026
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dissonance

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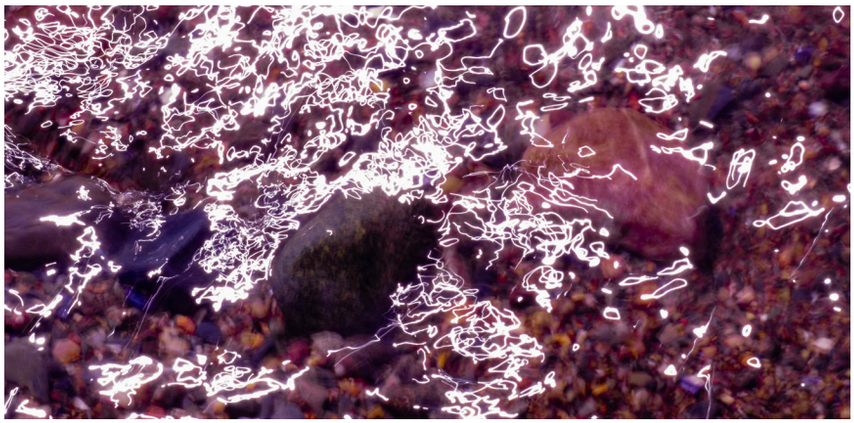

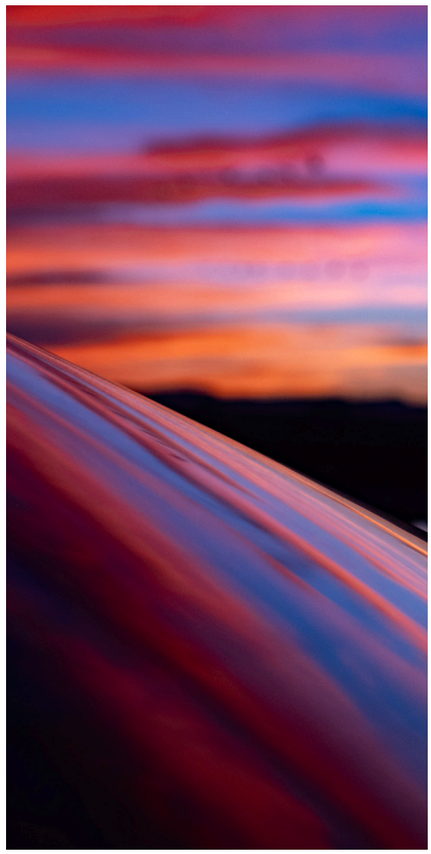

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LETTER FROM THE EDITOR

Bathing in the spotlight, a choir dances across the clefs as they turn to the final page of their piece. They pause on the penultimate note, two semitones lingering between the altos and the tenors. The audience leans toward the stage and seconds pass like hours, until they rejoin at the harmony and the crowd exhales. The echoing vocals fade, and the song concludes.

The euphonious sound of voices in sync is contrasted by the strained, dissonant lead-in. Just as every gold medal involves a strenuous race, and behind every perfect grade lies hours of studying, tension is a prerequisite for relief and release.

Dissonance is uncomfortable, but inevitable. As global access to education, international news, and social media improves, clashing opinions are formed and shared with more influence than ever before. We are pushed to face the sound of disagreement — to empathize with other points of view and prioritize tolerance, open-mindedness, and compromise.

Perfect harmony on such a scale is impossible, and would make for a boring and stagnant population. Oscillations between harmony and dissonance create music, in parallel to the way conflict and collaboration intersect to support innovation.

From the depths of Earth's oceans to Saturn's remote moons, this issue covers scientific exploration across our vast universe. Our writers tackle antibiotic resistance, ancient ship remains, house music, giant squids, and everything in between. Flexible and willing to challenge the status quo, scientists harness tension, failure, disagreement, and obstacles into unified data and discovery. Debate is a critical process when accuracy must overcome ego, and scientists are constantly working to improve and disprove each other in search of the truth.

I would like to thank our editors, writers, designers, photographers, web team, and e-board, who welcome dissonance with open arms to create unique ideas and grow the magazine. As an opposing factor to agreement and harmony, dissonance holds a quiet yet astounding value across forums and industries, creating space for the discussions and innovations we cover at NU Sci.

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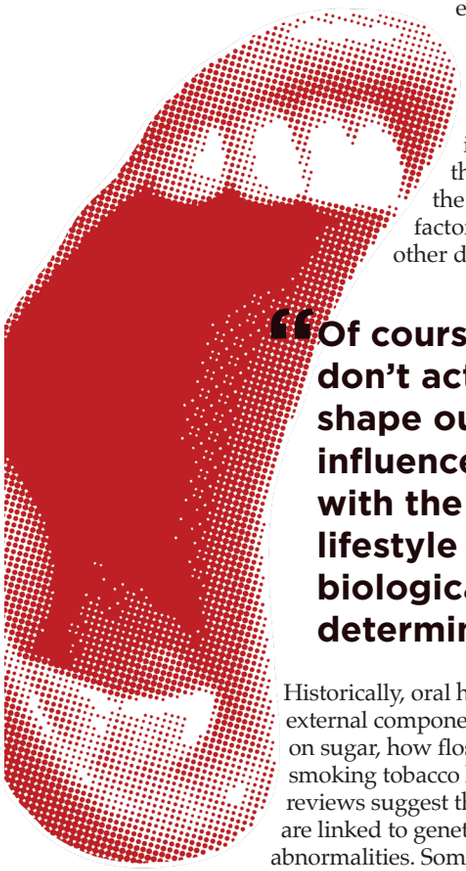
THE ROOT OF THE MATTER IS GENETIC

BY EMILY XU, BEHAVIORAL NEUROSCIENCE, 2026

DESIGN BY ANJANA BALAKRISHNAN, STUDIO ART, 2027

While diet, hygiene, and access to dental care all influence oral health, modern biology research finds that genes also play a crucial role in shaping everything from susceptibility to cavities and gum disease to oral anatomy (and even oral hygiene behaviors!). As the study of genetics advances to explore gene expression mechanisms, scientists are beginning to understand the function of genetic diversity in illness and development. Of course, genetics don't act in isolation to shape our mouths; they influence interactions with the environment and lifestyle factors to set the biological stage for other determinants to play out.



"Of course, genetics don't act in isolation to shape our mouths; they influence our interaction with the environment and lifestyle factors to set the biological stage for other determinants to play out."

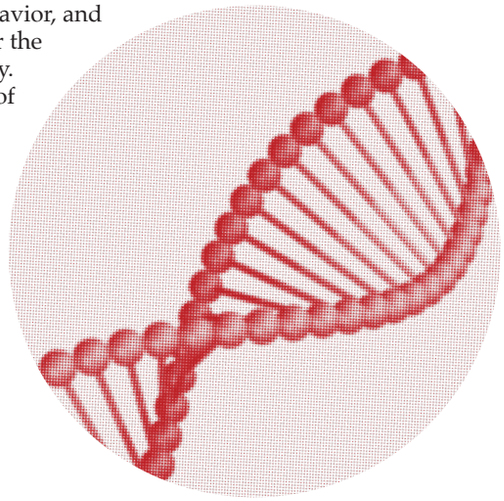
Historically, oral health research has focused on external components, such as how bacteria feed on sugar, how flossing removes plaque, and how smoking tobacco harms gums. However, recent reviews suggest that many oral health conditions are linked to genetic mutations and chromosomal abnormalities. Some of these common conditions include dental caries, also known as cavities, and periodontal disease. Their diagnoses are reliant on hereditary factors that increase susceptibility to decay and inflammation, with genetic variants that affect enamel structure, saliva composition, and immune responses. Not to mention, oral microbiomes can be informed by gene variants involved in carbohydrate metabolism and mucosal immunity. As a result, two patients may present with similar hygiene habits, but drastically different clinical outcomes.

Genetic determinants are also seen to strongly shape oral form and function through facial proportions, tooth size and shape, and jaw structure. Mutations in genes involved in oral development can severely alter tooth development. For example, PAX9 and MSX1 are transcription factors that regulate early tooth development, where loss-of-function mutants of these genes cause tooth agenesis, or missing teeth, by impairing initial development and disrupting epithelial-mesenchymal interactions for tooth formation. Interestingly, the condition is also associated with other syndromes such as Down syndrome and cleft lip and palate, suggesting an autosomal dominant inheritance pattern.

Genetic influence is not only limited to structure and disease, but can also inform behavioral and care-related outcomes. A recent study highlights that patients with genetic syndromes — such as Down, Prader-Willi, and Silver-Russel syndromes — show different patterns of oral hygiene behaviors compared to peers without. These differences reflect how the phenotypic expression of genetic disorders influences patients' ability to perform effective oral hygiene. For example, intellectual disability, motor impairment, sensory sensitivities, and oral structures contribute to the inaccessibility of preventative dental care. This illustrates how genetics shape the behavioral realities that alter the physical and cognitive bandwidths for individuals to execute oral health care behaviors.

Future practices may incorporate personalized genetic risk assessments that improve patient outcomes. For one, genetic testing could help clinically identify individuals who are at higher risk for cavities or periodontal disease, thus targeting the prevention of disease. Integrating biology with oral health can reduce stigma around disease, as diversity in outcomes is not based on "good" or "bad" habits, but rooted in complex gene and environment interactions. Additionally, risk assessments to identify genetic syndromes may help dentists tailor treatment plans for patients with disabilities to overcome communication barriers, technical challenges, and even enable accessibility to oral hygiene practices.

Understanding that oral health lies at the intersection of genes, environment, behavior, and clinical care is pivotal for the advancement of dentistry. By framing a landscape of risk and resilience in the mouth, research reveals an opportunity (and challenge) to apply interdisciplinary genetic research to improve clinical care practices.



IMAGES VIA ADOBE STOCK

Nature (2026). DOI:10.1038/s41586-025-10037-7
J Pharm Bioallied Sci (2025). DOI: 10.4103/jpbs.jpbs_1758_24
Dentistry Journal (2025). DOI: 10.3390/dj13030133
Scientific Reports (2024). DOI: 10.1038/s41598-024-80922-0

THE SCIENCE OF SCENT

BY NABIA CRAWFORD, CHEMISTRY, 2028

DESIGN BY KAVINA MANSUKHANI, COMPUTER SCIENCE AND DESIGN, 2028

One might imagine the smell of a mummy to be spoiled, rotting, or somewhat deathly. Surprisingly, experts describe the smell as pleasant, even sweet. But why are people smelling mummies in the first place? The answer lies in history, and what we're missing from it. It turns out that the past smelled completely different than our world today — the scents of turpentine, myrrh, and frankincense bled from Roman funeral rituals into cities, and Egyptian temples filled nightly with kyphi smoke, a blend of wine, myrrh, and juniper. Medieval homes scattered lavender underfoot to combat the underdeveloped sewage systems, and hippopotamus and crocodile fats were melted into oils and combined with cinnamon to create ancient perfumes. These mixtures of scents, amongst many others, have been lost to time as cultural practices have changed, the environment has shifted, and people have evolved.

A group of researchers is trying to change that. Matija Strlič, the lead scientist of the Heritage Science Laboratory in Slovenia, has focused his work on preserving and restoring historically significant scents. Odors have strong ties to psychology, influencing memories and learning, and shaping how we engage with the past.

> **"Reconstructed scents offer an opportunity to enhance gallery exhibits and the museum experience, as well as offering important information about conservation and restoration strategies for heritage objects."**

Museums often hold a sterile scent and present history as odorless when that was not in fact the case. This misrepresentation would be unacceptable regarding other important senses such as sight, touch, and hearing. Reconstructed scents offer an opportunity to enhance gallery exhibits and the museum experience. In fact, when the Prado Museum in Madrid installed a set of scents alongside a painting titled The Sense of Smell, viewers stayed for an astonishing 13 minutes, compared to the previous average of 32 seconds.

The identification and restoration process isn't as simple as just smelling and describing, though. A large part of how an olfactory profile — the structured composition of notes that defines a fragrance — is developed relies on the identification of compounds contributing to a smell. Analytical chemistry techniques, including gas chromatography and mass spectrometry, are often used to name scent-producing compounds within a larger profile. In the case of mummies, panel evaluations by olfactory experts were also conducted, with many describing the bodies with spicy, sweet, and woody odors. Some were more floral and herbal, while others were more citrus or tea-like. The difference in scents between different bodies offers a perspective into the respective history of each mummy — what oils were used to preserve the bodies, and what chemicals they were treated with.

Mummies aren't the only scents being analyzed. Strlič and his team were called to St. Paul's Library in London to document the library's distinct and beloved smell before scheduled renovations. Their work was documented and published such that any chemist could theoretically bring back the smell of the library in full force.
Another separate heritage project named Odeuropa aims to recreate smells that haven't been preserved at all. Amongst them are the scents of the Battle of Waterloo and 17th-century Amsterdam canals. These recreations rely on anything from documented building materials to details from old sermons. In this sense, the science of scent aims to improve preservation and conservation in the historical and archaeological world, improving our understanding of the past with each analysis.

Journal of the American Chemical Society (2025). DOI: 10.1021/jacs.4c15769
Thersites (2019) DOI: 10.34679/thersites.vol9.120
International Journal of History and Cultural Studies (2020). DOI: https://doi.org/10.20431/2454-7654.0604003
GMS Hygiene and Infection Control (2025). DOI: 10.3205/dgkh000550
Brain Sciences (2016). DOI: 10.3390/brainsci6030022

WINNING THE *ANTIBIOTIC ARMS RACE*

The creation of a novel tool to eliminate antibiotic resistance

BY LAILA COPELAND, CELL & MOLECULAR BIOLOGY, 2029

DESIGN BY ANJANA BALAKRISHNAN, STUDIO ART, 2027

Since the discovery of penicillin in 1928, scientists have been running a never-ending race against evolution. Every time patients are treated with antibiotics and antifungals for infections from strep throat to athlete's foot, the bacteria are exposed to new selective pressures. They, like any life form following Darwinian evolutionary patterns, must adapt to survive — and they do so by developing antibiotic resistance.

Each year in the United States, more than 2.8 million antimicrobial-resistant infections occur, leading to at least 35,000 deaths. At least 700,000 people die each year because their diseases cannot be treated with antibiotics. This means new antibiotics must be discovered, isolated, developed, tested, and mass-produced — and the race against evolution continues.

> **"At least 700,000 people die each year because their diseases cannot be treated with antibiotics. This means new antibiotics must be discovered, isolated, developed, tested, and mass-produced — and the race against evolution continues."**

Germs — which are microscopic organisms like fungi, bacteria, and protozoa — are found everywhere in our bodies, but some cause more harm than help. Antibiotics are medications administered to kill these germs. Penicillin, for example, destroys bacteria by rupturing their cell walls. However, bacteria can evolve to become resistant to penicillin by acquiring an antibacterial resistance (AR) gene from a resistant neighbor. This is a process called horizontal gene transfer, when genes are not passed down through generations like in humans, but instead are picked up from other bacteria that have survived nonlethal doses of antibiotics.

Once the antibacterial resistance gene, encoded on a small DNA molecule called a plasmid, is picked up, it can easily be integrated into the bacteria's genome. However, two professors at the University of California San Diego, Ethan Bier and Justin Meyer, have found a way to harness gene-editing technology to neutralize these plasmids.

The saga started seven years ago, in 2019, when Bier and his team created a tool called Pro-Active Genetics (Pro-AG), which allowed them to genetically inactivate the AR genes within individual bacterial cells using CRISPR Cas-9.

CRISPR Cas-9 is a gene editing tool found in bacterial immune systems. It uses the Cas9 protein as molecular scissors, guided by a template sequence to create a double-strand break in DNA at a specific genetic locus. The natural repair mechanisms of the cell then stick the broken ends of DNA back together, often disrupting or deleting the target gene sequence.

Bier's Pro-AG tool used CRISPR-Cas9 to successfully cut the target AR gene, and inserted a new sequence that inactivated all copies of the AR gene throughout the bacteria's genome. Unfortunately, this strategy was clunky.

Pro-AG involved three separate plasmids to encode the CRISPR-Cas9 machinery, and there was no way to make bacteria spread these specific plasmids using horizontal gene transfer. A new delivery method was needed.

In 2023, scientists at the University of Exeter engineered a single plasmid-based delivery vehicle that delivered a CRISPR Cas-9 complex designed to target AR sequences in the bacterial genome of *E. coli*. Like Pro-AG, this plasmid not only cut pre-existing AR genes, but could also block future uptake of AR plasmids via insertion of a short DNA sequence. This plasmid was then easily copied between bacteria using horizontal gene transfer, which inactivated their AR genes.

In essence, scientists turned the bacteria's own gene-sharing ability against it. By enabling the spread of an engineered plasmid between bacteria, Bier and his team could take advantage of the way bacteria often coagulate in biofilms, where they use the power of teamwork to resist antibiotics.

Biofilms can grow on a wide range of media, from medical devices like catheters and heart valves to sewage treatment plants to hot springs. While these communities of microorganisms contribute in their own way to our ecosystem, it's not in our best interest to host them inside our bodies. Biofilm-based infections can evade the immune response and have higher immunity to antibiotics, often associated with chronic infections like cystic fibrosis.

Bier wanted to find a way to take advantage of the way bacteria within biofilms can pass around AR genes in order to distribute Pro-AG. He was joined by Meyer, an evolutionary microbiologist, to combine their Pro-AG methodology with the CRISPR-Cas9 plasmid, which could theoretically penetrate these dense microbial communities. In early 2026, the pair revealed that they had engineered Pro-AG into a single plasmid that contained everything necessary for its dissemination into AR-positive bacteria, naming it pPro–MobV. Bier claims that due to the effectiveness of large-scale administration of pPro-MobV, scientists are able to "take a few cells and let them go to neutralize AR in a large target population."

However, there are still concerns about pPro-MobV's level of regulation. If horizontal gene transfer spins out of control, it poses risks of disrupting microbial ecosystems and unintended consequences in environmental bacteria. Self-spreading genetic tools like this one always raise some level of concern, which is why long-term monitoring would be necessary.

On the bright side, the creation of a self-propagating AR inactivation system could quell fears about 'superbugs' rendering current antibiotics useless. Meyer claims their technology is able to "actively reduce the spread of antibiotic-resistant genes." It could lower the rate of hospital-acquired infections, and help protect certain strains of wheat. Bier and Meyer's pPro-MobV requires further monitoring, but if successful, would be a huge boon in the century-long race against antibiotic resistance.

Nature Antimicrobials and Resistance (2026). DOI: 10.1038/s44259-026-00181-z
BMC Microbiology (2023). DOI: 10.1186/s12866-023-02954-2
Microbiology (2023). DOI: 10.1099/mic.0.001334
Nature Communications (2019). DOI: 10.1038/s41467-019-13649-6
Nature Reviews Microbiology (2014). DOI: 10.1038/nrmicro3270

PHOTOS VIA ADOBE STOCK

PERSISTER CELLS

AGENTS OF ANTIBIOTIC TOLERANCE & CHRONIC INFECTION

BY CJ CROMBIE, CELL & MOLECULAR BIOLOGY, 2026
DESIGN BY EDNA AMAN, CELL & MOLECULAR BIOLOGY, 2029

Deep in the throes of infection, bacterial footsoldiers wage war against human cells, invading the tissues of the body. They multiply, stealing nutrients from their host and secreting offensive toxins. The host immune system rages in response: phagocytic cells swallow bacteria whole, the body's temperature rises, and tissues become inflamed. The immune response is so effective that it will often kill the host in the process of eliminating an infection, causing sepsis, shock, and organ failure.

Medicine met this challenge with penicillin, the first drug to target and kill pathogenic (disease-causing) bacteria inside of a human host. Penicillin works by binding and inactivating key proteins needed for synthesis of the bacterial cell wall. This causes the swiftly-multiplying bacteria to lyse, or "burst," and die. Human cells lack a cell wall, so the drug can target a pathogen without toxic effects on the infected host.

The concept of antibiotics is relatively straightforward: identifying a target (usually a protein) involved in a necessary bacterial function, then administering a drug which halts these functions. This molecular target should only be present in the bacteria; otherwise, human cells are at risk. In the plethora of metabolic processes undertaken by bacteria, there are many components that can serve as drug targets. For years after the discovery of penicillin, antibiotics entered a golden era of sorts. More and more drug classes were discovered, and they were increasingly effective in eliminating infections. Inhibitors of protein synthesis, DNA replication, cell wall formation, and other metabolic processes entered the clinic and stayed there.

"Around the end of the 20th century, the discovery of new antibiotics slowed. Rising to counter this was a silent pandemic, slowly taking the world by storm."

Around the end of the 20th century, the discovery of new antibiotics slowed. Rising to counter this was a silent pandemic, slowly taking the world by storm. Antibiotic resistance had begun to develop.

The misclassification of antibiotics as "magic drugs" expanded their use to agricultural settings and non-bacterial medical applications like viral infections. Bacteria—because they divide so quickly—mutate frequently. When antibiotics aren't administered at a high enough concentration to eradicate entire bacterial populations, these organisms adapt to the stressful environment. Selective pressure ensures the survival of helpful mutations that render the antibiotic ineffective. These mutant cells are called resistors and comprise 1/6th of all bacterial infections confirmed in medical laboratories.

Resistors are characterized by an immense ability to adapt. This ability is enhanced by improper antibiotic dosing and poor patient compliance: administering too little antibiotic or not finishing those nasty-tasting pills once symptoms begin to improve. This kills some bacteria and allows the survivors to re-populate and spread their beneficial mutations against antibiotics. Bacterial infections are once again becoming the formidable threat they were prior to penicillin.

There is a second, lesser-known component to the resistance problem. Persisters are bacterial cells distinctly different from resistors but similarly unaffected by antibiotics. The persister phenomenon was first conceptualized in the 1940's by Dr. Joseph Bigger, who identified this sub-population of cells as a troublesome component of bacterial infections. Where resistant cells have developed some active mechanism against the drugs, persister cells are bacteria that enter a period of dormancy, either randomly or due to the presence of an environmental stressor like nitrogen depletion or an antibiotic drug.

The dormant period of persisters is similar to the bacterial growth stage known as the stationary phase, where metabolic processes slow and cell division halts. This results in bacterial tolerance to drugs, where they mitigate the

effects of an antibiotic drug not by actively adapting to it, but by halting the processes the drug targets. Important to note is that resistance is heritable, while persistence is not. This is because resistance results from a genetic alteration that can be passed on, while persister cells do not experience a permanent change to their DNA.

The key to persistence is the ability to "re-awaken" from this period of dormancy. In the presence of a stressor, they remain unaffected, but the moment the stressing factor is removed, persisters can resume normal metabolic activity. This allows the bacteria to repopulate and jumpstart the infection process anew in a phenomenon known as recalcitrance. Recalcitrance is a significant contributor to chronic and recurring infections like UTIs, tuberculosis, and cystic fibrosis.

The Lewis Lab at Northeastern University studies persister cells responsible for tolerance to antibiotics. So far, the lab has identified three significant mechanisms of persister formation among various bacterial species: toxin-antitoxin modules in Escherichia coli (E. coli) and nitrogen depletion as well as random fluctuations in ATP levels in both E. coli and Staphylococcus aureus (S. aureus).

Toxin-antitoxin modules are two-component systems employed by many bacteria, and they provide a unique self defense mechanism. A toxin, such as tisB in E. coli, is released to protect the cell from a threat, briefly halting cell processes, then is later neutralized with an antitoxin molecule. Antitoxins are highly unstable and susceptible to degradation, meaning many E. coli can remain in this state of metabolic pause without neutralizing the toxins. Further, when the toxin halts bacterial growth, it removes the targets for an antibiotic, and again, persisters are born. Notably, toxin-antitoxin modules have not been found to play a role in the formation of S. aureus persisters.

Persisters can also form when bacterial cells lack nitrogen. Nitrogen is a necessary component for a variety of bacterial metabolic processes, including the synthesis of DNA, proteins, and other vital macromolecules. However, it is also a limited environmental resource for cells, and as such, many bacterial species compete for available nitrogen. E. coli and S. aureus, among many other bacterial species, have adapted a stress response that they can employ in the absence of nitrogen ("nitrogen starvation"), and this stress response is believed to increase the proportion of cells that become persisters. This occurs through a cascade of intracellular signals, which slows metabolic activity and results in persister formation.

Finally, the lab has observed that low ATP is associated with an increased formation of persister cells. Bacterial cells often randomly ("stochastically") enter the stationary phase, where intracellular ATP decreases. This low-energy state slows metabolic activity, decreasing the activity of antibiotic targets and producing a tolerant population (persisters).

The Lewis Lab also aims to identify potential solutions to the problem of persistence by supporting natural drug discovery efforts to address both resistance and persistence. In 2013, Principal Investigator Kim Lewis discovered that an old molecule called Acyldepsipeptide (ADEP) could eliminate persister cells both in vitro and in vivo (mouse infection models). ADEP targets a protease called ClpP, hijacking it to degrade mature proteins in the cell. Normally, ClpP only degrades misfolded or damaged proteins, so this corruption is devastating for bacteria.

Beyond small molecule discovery efforts, the Lewis Lab has also pursued more creative endeavors such as pulse-dosing and metabolic stimulation. Pulse-dosing (another contribution of Dr. Bigger) involves treating an infection with multiple rounds of antibiotics, killing off the active population and leaving persisters behind. After each treatment, the antibiotic is cleared from the system and persisters are allowed to re-awaken in the absence of a stressful environment, and are then hit again with the drug. This is repeated until, in theory, all persisters are eradicated. Metabolic stimulation involves supplementing one step in the bacterial citric acid cycle in an effort to restore metabolic function, followed by antibiotic treatment to kill newly-awakened cells.

"Resistance dominates public perception of the crisis of antibiotic inefficacy, but the Lewis lab has demonstrated the clear threat posed by tolerant persister cells."

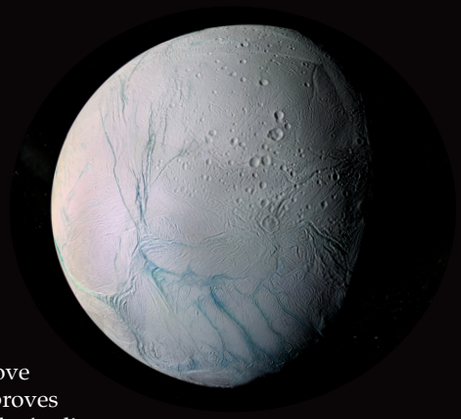
Resistance dominates public perception of the crisis of antibiotic inefficacy, but the Lewis lab has demonstrated the clear threat posed by tolerant persister cells. It isn't quite clear what the future of persister cells holds, but without the research of microbiologists around the globe, we may continue to observe an increase in the frequency, duration, and severity of bacterial infections. What is obvious is that the solution will require scientific efforts as persistent as these little cells themselves.

Nature Portfolio (Preprint). DOI: doi.org/10.21203/rs.3.rs-7237366/v1
PLOS Biol (2021). DOI: 10.1371/journal.pbio.3001194
Molecular Cell (2018). DOI: 10.1016/j.molcel.2018.01.003
Nature (2013). DOI: 10.1038/nature12790
PLOS Biol (2011). DOI: 10.1371/journal.ppat.1002121
PLOS Biol (2010). DOI: 10.1371/journal.pbio.1000317

PHOTO VIA ADOBE STOCK

BEYOND THE ICE WALL

Modeling the Dark Chemistry of Enceladus



BY BAVESH MATAPATHI, MECHANICAL ENGINEERING, COMPUTER SCIENCE & PHYSICS, 2026 & JIAJIA FU, BIOENGINEERING, 2026

DESIGN BY CLAIRE HANSON, HEALTH SCIENCE & ENTREPENURSHIP, 2027

Our solar system is home to many worlds with rich chemical processes. While we consider Earth to be the crown jewel of complex biochemistry, we are now equipped with technology to peer into worlds beyond our own. Among them, orbiting Saturn, is Enceladus — a moon hiding secrets underneath its thick icy exterior.

In 2005, the Cassini spacecraft discovered that plumes of water vapor were spraying into space from huge fissures in Enceladus' surface at its south pole. This, along with the detection of silica crystals, supports the theory that the 310-mile-wide moon hosts a large subsurface ocean providing water for the plumes. Furthermore, it provides a model of Enceladus as hydrothermally active — burbling salty waters churned with Saturn's gravity.

Cassini's Cosmic Dust Analyzer (CDA) recorded hundreds of thousands of ice-grains in the E-ring, a diffuse cloud beyond Saturn's visible rings, where some of the ejecta from the plume could settle for days to decades. Charged particles, trapped in the magnetosphere of the host planet, bombard the ice grains, instigating chemical reactions outside of Enceladus. This data is crucial for scientists mapping potentially organic chemical activity that is endogenous to ice-shielded subsurface oceans.

A 2025 study from Cassini's fly-by of Enceladus (E5) sampled ice-grains directly from the plumes. This ensures that the ice-grains analyzed are freshly collected, thereby increasing certainty that the compounds detected arise from the subsurface ocean rather than Saturn's E ring prone to external radiation. The fly-by of plumes also had a high-encounter speed, providing higher resolution by reducing interference from water-clusters that can mask signals from organic species.

Through mass spectrometry, an analytical method that measures ion mass ratios to classify compounds, the researchers identified the presence of aromatic groups. These highly stable, carbon-ring species are a type of hydrocarbon that provides the backbone for all known life. Single-ring aromatics play a pivotal role in the organic chemistry on Earth's sea floor, and serve as precursors for more complex organic compounds synthesized by microorganisms and animals. Although detecting these compounds does not prove biological activity, it improves the likelihood of Enceladus' saline oceans for harboring organic reactions.

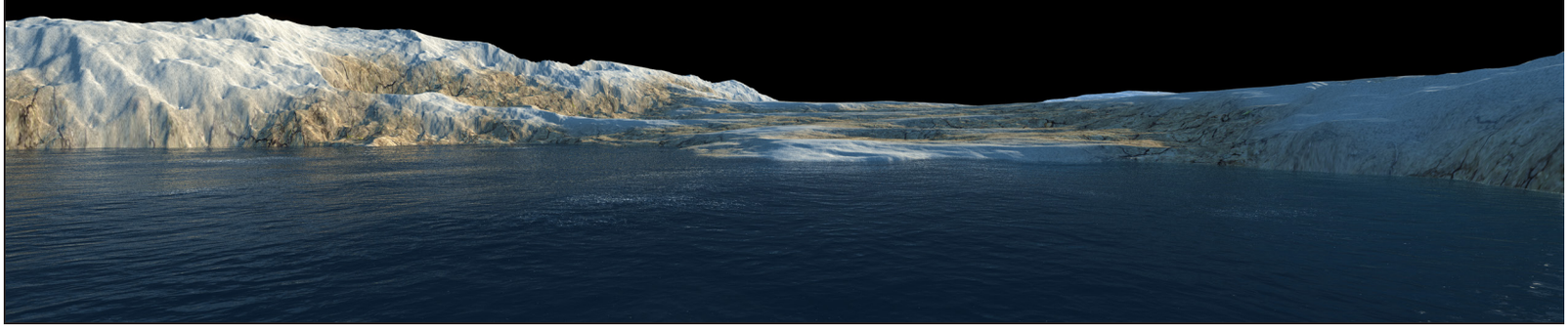
Fragments discovered from the ice-grains now open a possibility for a vast prebiotic chemical reaction network in the subsurface ocean. Additional organic hydrocarbons are modeled, including aldehydes, esters, and ethers, which further act as links and bridges for biomolecule formation. Evidence of this can be seen with the detection of more complex fragments, including derivatives of maleic acid. Researchers theorize that reactions interacting with the nitrogen and oxygen-rich, warm saline oceans of Enceladus could allow the synthesis of amino acids and lipids, which are crucial for metabolism.

While the fragments found in the plumes have piqued the interest of scientists studying subsurface organic chemical dynamics, new research presented recently at a planetary science conference in Finland shows that the organics from the plumes can also originate from the Enceladus surface. An experiment conducted by a team of researchers from Italy's National Institute for Astrophysics showed that a recreated surface of Enceladus produces carbon monoxide, cyanide, and ammonium when bombarded with charged particles. As these samples were gently warmed, more complex organics appeared, which served as precursors to amino acids. This research complicates the belief that biochemical activity of Enceladus occurs purely in its oceans.

Nonetheless, scientists are still excited about alternative pathways of organic chemistry occurring on Enceledus, raising interest for future missions by ESA and NASA, proposed to arrive in the 2050s. The aim would be to have a lander on the surface to sample fresh ice near the plumes. Enceladus and other bodies of interest in our solar backyard provide laboratories to study complex biochemistry and consider the possibility of life outside our world as we inch towards considering the habitability of new worlds.

Nature Astronomy (2025). DOI: https://doi.org/10.1038/s41550-025-02655-y

PHOTO VIA ADOBESTOCK



ALPHA GENOME

The age of artificial intelligence in molecular biology

BY RAINA LANGLAIS, CELL & MOLECULAR BIOLOGY, 2029
DESIGN BY CLAIRE HANSON, HEALTH SCIENCE AND ENTREPRENEURSHIP, 2027

It is no longer an understatement that artificial intelligence has infiltrated everyday life. Now, it is being applied to one of the most challenging frontiers of biology: interpreting the regulatory genome. The human genome consists of over three billion base pairs, and only 1-2% of it encodes proteins. The remaining 98% is responsible for regulating gene expression and other vital biological functions. Google DeepMind has recently unveiled AlphaGenome, a deep learning model that is designed to help scientists identify how genetic variation disrupts the complicated networks controlling gene expression.

The human genome is frequently described as an "instruction manual for life," and for good reason. It carries the molecular blueprint that constructs and maintains the human body. This instruction manual is written in a four-letter chemical code, organized into approximately 20,000 to 25,000 genes that encode proteins and control a vast array of biological functions. However, these genes only make up a minute fraction of the genome. The vast majority of the genome involves regulatory DNA that governs the way in which these genes are expressed. Understanding how this complicated regulatory landscape operates and how a single variation in the genome can lead to disease is a problem that decades of advances in computational technology have not quite addressed yet. It is exactly this challenge that AlphaGenome seeks to address.

AlphaGenome is the work of Google DeepMind, a United Kingdom-based artificial intelligence research laboratory that is renowned for creating the Nobel Prize-Winning AlphaFold system, a highly sophisticated computer tool for building molecular structures and designs of genes. Their newest creation, "AlphaGenome," represents the latest extension of Google DeepMind's ambitious goal to utilize artificial intelligence to solve fundamental scientific inquiries.

So, how does it work? Unlike previous computational models, which only accept short pieces of DNA as input, AlphaGenome is capable of parsing up to one million letters of DNA. By analyzing the properties of genetic material, it can make thousands of predictions about gene expression level, regulatory elements, transcription factor binding, and chromatin accessibility. Garnering impressive speed and accuracy, AlphaGenome was trained on publicly available human and mouse genome databases, such as ENCODE, 4D Nucleome and FANTOM5. By combining long-context sequence modeling with multimodal prediction, AlphaGenome matches, and in some cases, outperforms earlier regulatory models. In addition, beyond computational benchmarking, the model allows scientists to move easily from data sequencing to more complicated applications like synthetic biology and disease processes. For instance, the model can predict disruptions in RNA splicing related to the onset of cystic fibrosis. Further, it can assist in the design of regulatory motifs that selectively turn on or suppress gene expression.

Despite the experimental success the model has had, Google DeepMind made sure to acknowledge its shortcomings. For instance, the model struggles to identify regulatory sequences further than 100,000 base pairs away, and it is unable to fully capture the dynamic, context-dependent nature of cells. Additionally, AlphaGenome was not designed to comprehensively explain how variants influence complex diseases and can only predict their potential regulatory effects.

Beyond its technical accomplishments, AlphaGenome also invokes the broader question concerning the role of artificial intelligence in science. Can these machines help us interpret biological systems too complex for traditional computational analyses alone? There is no question about AI ever being able to replace experimentation, but developments like AlphaGenome demonstrate that machine learning can serve as a powerful ally in decoding the complexities of science.

Nature (2026). DOI: 10.1038/s41586-025-10014-0
Nature (2025). DOI: 10.1038/d41586-025-01998-w

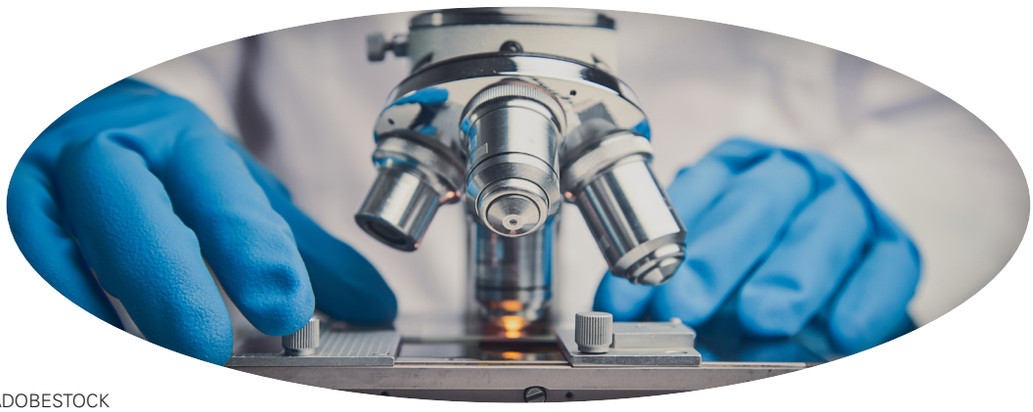


PHOTO VIA ADOBESTOCK

Science in Conversation with Justin Hayes

Going with your gut

BY ELLA HANNES, CELL & MOLECULAR BIOLOGY, 2028

DESIGN BY ELLA HANNES, CELL & MOLECULAR BIOLOGY, 2028

Academia or industry? This is a question any PhD candidate in life sciences has had to face. Will they join a university, collaborating to answer science's pressing questions, or work in private industry, bringing health innovations to market? While they used to be viewed as mutually exclusive fields, the lines between industry and academia are blurring. It's becoming increasingly accepted that the two are not separate; they're married. This is certainly true for PhD candidate and CEO Justin Hayes.

Currently pursuing his PhD in Chemical Engineering, Hayes has a collaborative research project shared between two labs at Northeastern University: the Koppes Lab and the Woolston Lab. He investigates hydrogen sulfide (H2S), a gas produced by bacteria in the gut, to answer a big question: Can H2S be harmful to gut health?

This question has plagued health scientists for many years because of a persisting problem; the amount of H2S in the gut cannot be reliably measured. Or rather, it couldn't, until Hayes went to work. By bioengineering variants of E. coli to produce precise levels of H2S, Hayes tests the effects of the gas concentration in a model gut developed in the Koppes Lab. His data shows that H2S reduced inflammation at low levels, but did the opposite at high levels, bringing an integral piece of the gut-health puzzle to the table.

Q: What drew you to your current, niche research topic: metabolites of the gut microbiome?
A: Coming into grad school, I thought I was going to do water treatment. It was really my driving factor, motivating factor is like, save human lives, reduce human suffering . . . I really liked the idea of the project that Dr. Koppes and Woolston put together. So in human health, you can actually impact probably more people with all the tools that are available now in medicine.

With a background in chemical engineering, Hayes was able to address this biological problem from a unique point of view. His understanding of principles like heat, reaction kinetics, and thermodynamics allowed him to attack the problem in a new way and develop it further into a product.

Q: Was there a point that your research shifted from "interesting basic science" to "this could actually be a therapy"? What was that transition like?
A: We had sort of always known there was the potential to make uh, you know, a microbial drug or a microbial product out of what we were doing . . . Once we demonstrated we could manipulate different molecules in the gut in animals . . . that was really a really big inflection point for us where it's like, okay, not only can we do this in a lab setting, we can do this in a living being. That was pretty exciting.

Hayes founded Concordance Therapeutics in 2024 to bring discoveries of the gut microbiome from the lab to the market. His scientific advisors supported his mission, and allowed him to adapt his research in the lab to coincide with his entrepreneurial goals. Additionally, he sought advice from industry professionals with their own start-up or managing experience.

> **"There's a lot of solutions that could change the world that have died, are dying, or will die because people just won't make the effort to translate that."**

Q: How did you prepare to start your company? Did you have any important supporters or mentors that helped you?
A: The first thing I did was the NSF I-Corps program sponsored at MIT. You basically come with an idea and they teach you some lean methodologies in the startup world. And the real focus of the program is just to go out and talk to people who may use your products or stakeholders within that realm . . . How do you prepare yourself to become CEO of a biotech company, raise millions of dollars, manage people—all that's an undertaking. And then there's probably dozens of people in my network that have given me advice, guided me over time. So if people are interested in doing this, it's having a great network of people that are just willing to help.

Since its founding, Concordance Therapeutics has grown rapidly, taking first place in the Boston Shark Night competition and moving operations into The Engine labs in Cambridge. Hayes' success has depended on his ability to balance the two worlds of graduate school and biotech industry.

Q: Now you're juggling being both a PhD student and running your company. What does that balance look like?
A: Fortunately for me, the science that I'm doing for my thesis is very aligned with the company. So in that sense, somewhat two birds, one stone. But I try to plan

my research very efficiently . . . And then for the startup, it's a lot of phone calls, talking to people, working on small business grants or talking to investors or accelerator programs or whatever, learning, doing something, building out our data room. So it's just a lot of computer work. So that actually makes it a little bit more flexible.

For Hayes, his role as CEO will become a full-time job after graduation. With the company taking off, investments climbing, and his products coming closer to market, it's easy to forget the moments when the stars didn't align — when an experiment failed, when funding fell short, or when doubt crept in.

Q: Was there ever a moment when you thought "I'm not cut out for this" or "it's not going to work"? What happened?
A: There's a lot of imposter syndrome. People experience that all the time, whether it's in research or in the startup world. So it's something you've got to deal with. There's so much uncertainty. Sometimes you're uncertain about yourself. Sometimes you're uncertain about the market, the research, or whatever . . . As long as you stay alive as the company, you just wait for that golden egg to appear and save the day . . . I think there's a phrase 'default alive'—as long as you can stay alive, as the founder you just kind of have to have that mentality. It's like no matter how bad today is, just stay alive until the next one and just keep going.

Staying both persistent with current ideas and flexible enough to pivot is a delicate balance that defines both research and entrepreneurship. It's a lesson learned through trial and error; the kind of hard-won insight that can't be taught in a classroom. When asked what advice he'd give to students hoping to follow in his footsteps, Hayes' answer was characteristically practical.

Q: What's something you wish you knew before you started your company?
A: Talk to as many people as you can . . . Reach out to those people who in your mind are like superstars and you'd think they'd never talk to you. But they will. Go talk to those people. Go talk to the former CEO who exited for billions of dollars . . . There's just so much to know. I mean, there's so much I don't know right now. And I'm going to learn more today and tomorrow.

Q: Do you have any advice for students who want to pursue both the research and entrepreneurship paths?
A: Find people who have done it before. Talk to them. Give yourself time too . . . Think about it maybe in your second or third year. Not necessarily that you need to have everything figured out, but like, 'Hey, I'm interested in just exploring the idea.'

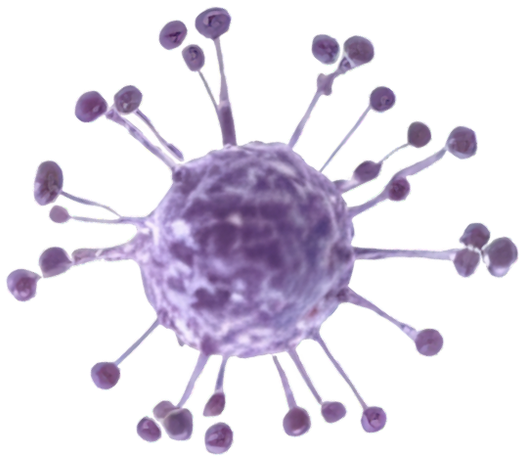
The recurring theme of learning from others reflects the truth that great accomplishments cannot be achieved alone. Some of the greatest medicinal discoveries came from collaboration; COVID-19 mRNA vaccines were developed at the University of Pennsylvania and brought to patients with the help of Pfizer, Jennifer Doudna developed CRISP-Cas9 gene editing at UC Berkeley and co-founded multiple companies to market her innovations as genetic therapies. Yet, bridging the gap between academic discovery and mainstream medicine remains a challenge.

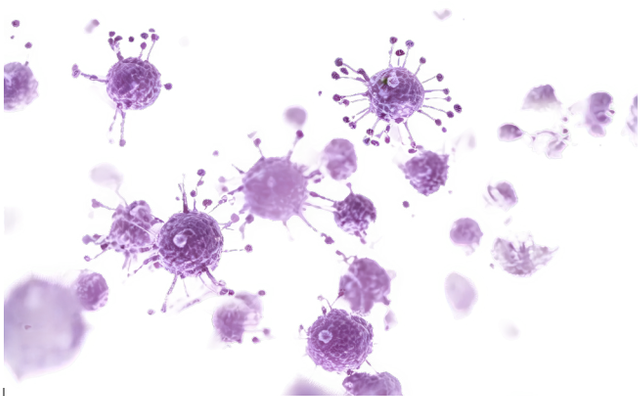
> **"Talk to as many people as you can . . . Reach out to those people who in your mind are like superstars and you'd think they'd never talk to you. But they will."**

"I personally think that scientists should take some of that burden on themselves and think like, 'Hey, not only do I have to do the science, but I actually have to go the next step and go bring this to market.' Because I think there's so many ideas that die in academia. Scientists — we always say we're going to change the world — but I think there's a lot of solutions that could change the world that have died, are dying, or will die because people just won't make the effort to translate that," said Hayes.

It's impossible to know how many great ideas never made it off the lab bench. But scientists and entrepreneurs like Hayes are taking the burden upon themselves, now more than ever, to fulfill the true purpose of science: developing novel discoveries to benefit others.

Read Hayes' full interview on the NU SCI Magazine website.





PHOTOS VIA ADOBE STOCK

BRINICLES

The "life-giving" harbingers of death

BY MIKAYLA TSAI, BEHAVIORAL NEUROSCIENCE, 2026

DESIGN BY ANJANA BALAKRISHNAN, STUDIO ART, 2027

Numerous resilient forms of marine life reside on the bottom of the ocean — whether that be worms, brachiopods, or bivalves. Due to the development of unique adaptations in the face of stressors, these sedentary organisms remain steadfast amidst harsh conditions of low food availability, high pressure, and tempestuous waves that pound overhead. While each species of organism operates differently, the "ice finger of death" does not discriminate between starfish and crayfish. Upon contact, the hubbub of ecological interactions is reduced to eerie stillness, and all life is snuffed out.

> **"Upon contact, the hubbub of ecological interactions is reduced to eerie stillness, and all life is snuffed out."**

When water rich in salt content leaks out of sea ice, this phenomenon can create tendrils of ice called a brinicle. While these structures reach extremely low temperatures dropping below -20 degrees Celsius, their formation is similar in mechanism to how a hydrothermal vent is created, just in reverse. At the depths of the ocean, these vents exist along the sea floor, which expel chemical-rich boiling water upward. In contrast, brinicles are known to release supercooled, highly salinated water in the downward direction. During the brinicles' descent into the deep, a hollow growing tube is created, which freezes the surrounding air.

While brinicles have a capability of snuffing out the livelihoods of slow-moving organisms alike, they also have "life-giving" qualities which are nothing short of paradoxical.

As the brine is driven out from the ice pockets, the drastic and sudden switch to frigid temperatures can cause variations in acidity, which can drive meaningful chemical reactions. In fact, studies have shown that the brine-rich compartments in which brinicles originate carry high concentrations of chemical compounds, fats, and lipids.

"They can act as a primitive membrane — one of the conditions necessary for life," noted Bruno Escribano, a researcher at the Basque Center for Applied Mathematics.

In particular, sea ice diatoms are a type of single-celled marine microalgae that use nutrient-rich brine channels to migrate closer to the light and maximize their ability to photosynthesize. In comparison to the surrounding seawater, these saltwater channels are more capable of delivering high concentrations of salt ions and macronutrients. During periods of warmer temperatures, diatoms largely comprise "phytoplankton blooms" of marine algae that spread rapidly. In turn, this phenomenon serves as a crucial food source for smaller organisms — consequently sustaining the broader food web and contributing to ecosystem stability.

Studies have also shown that the formation of brinicles may not only exist below the tides, but on extraterrestrial bodies, such as Europa, an icy moon with an environment that carries a high potential for habitability, and likely contains twice as much water as the Earth's oceans combined. Above the water's surface is an "ice shell" which is estimated to be around 10 to 15 miles in thickness. Consequently, Europa's ice shell may drive the formation of brinicles to serve as a channel in which space-related materials are delivered into the underlying ocean.

While further research is still needed, these saltwater sinks reveal a novel pathway into how biological life may originate — all from a "primordial soup" of organic molecules that hold immense potential for something bigger.

Frontiers (2023). DOI: 10.3389/frym.2023.1046858
Sage Journals (2019). DOI: 10.1089/ast.2018.1826

PHOTO VIA ADOBE STOCK

MODEL OR MIND?

Defining humanity in the age of organoids

BY OLIVIA MULLER-JUEZ, BIOCHEMISTRY, 2028

DESIGN BY ANJANA BALAKRISHNAN, STUDIO ART, 2027

In 2011, a wayward cell culture revolutionized drug development and research. Dr. Madeline Lancaster, a cell biologist then based at t Dr. Madeline Lancaster, a cell biologist then based at the Institute of Molecular Biotechnology in Vienna, was working on culturing embryonic stem cells into circular neural cell arrangements when she noticed a botched batch acting strangely. As though following some omniscient signal, the cells joined together in synchronized agreement, slowly organizing into a pea-sized segment of neural circuitry: a lab-grown bit of brain.

Research conducted by Lancaster and her colleague Juergen Knoblich, as well as Yoshiki Sasasi at Kyoto University Graduate School of Medicine, identified the cause and applications of this phenomenon. By activating various signaling pathways with a cocktail of growth factors in a 3D extracellular matrix or media, scientists can manipulate either embryonic or induced pluripotent stem cells to generate cells that self-organize into functional tissue. The result is kernel-sized "organoids" reflective of tissue grown in a live human.

Today, lab-grown organoids substantiate entire areas of research, providing golden alternatives to animal testing and organ donation. Organoids have changed the game by facilitating disease modeling, toxic drug screening, and therapy development. Advancements in countless projects can be attributed to foundational organoid trials, including cystic fibrosis modeling, regenerative liver therapy, and anti-cancer drug response research.

Just as this technology reaches peak velocity, however, many are calling to pump the brakes and consider the impending future. Following the science's trajectory, it is possible we may one day be able to grow functional, living brains out of a dish. To what end can we utilize this organism and at what point do we care? Of all ethical borders science treads, the question of how to classify organoid brains poses a critical crossroads, threatening to expose the human insecurity of unremarkable consciousness and lineate domains between bodies and souls.

A meeting held in Northern California in January 2026 aimed to brainstorm contentions surrounding these issues. The event was hosted by Dr. Sergiu Pașca, a researcher from Stanford University investigating pain pathways in brain organoids, and attended by bioethicists, researchers, and journalists.

PHOTO VIA ADOBE STOCK

The panel covered an important question: at what point in organoid development would the specimen attain human capacities? In other words, when should one start treating the organoid as a being rather than a tool? This becomes a more pressing issue with the introduction of combination organoids, or assembloids. Assembloids connect organoids into more complex fractions of an organ, bringing them a step closer in appearance to living beings.

"Of all ethical borders science treads, the question of how to classify organoid brains poses a critical crossroads, threatening to expose the human insecurity of unremarkable consciousness and lineate domains between bodies and souls."

While the panel did not decide on a firm boundary of consciousness, they recognized essential awareness circuitry as a necessary indicator. Take Pasca's pain pathway research: as the assembloids are disconnected from a pathway for "emotional aversion," there is no potential for their suffering. The mere association of the neural signal pathway with pain does not mean the organoids are in pain. This highlights the incomplete nature of organoids, emphasizing that their disconnectedness and isolated existence safeguard them against suffering, or real consciousness.

With organoids still in the early stages of development, researchers are confident no ethical lines have yet been crossed. That said, the technology is rapidly improving; hopefully alongside our understanding of consciousness.

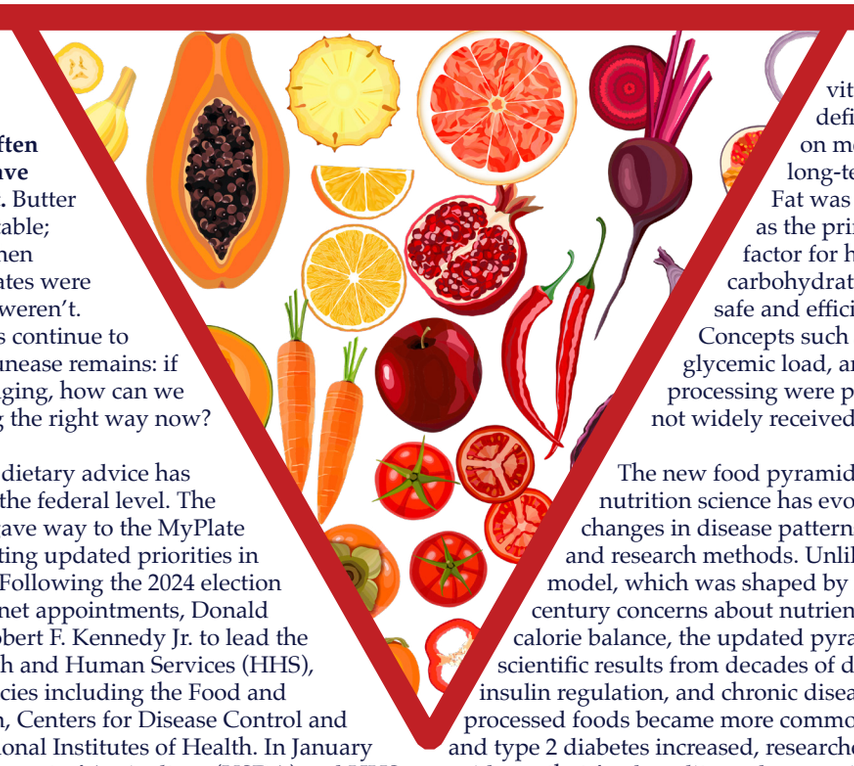
As the panel ended, there was a clear consensus over the need to keep the public informed. Public and private institutions in the US all demand certain regulations to reign in unfettered scientific ambition. Ultimately, however, as organoid technology improves and these issues become more urgent, it becomes a public responsibility to decide the ethical limitations of science.

Science Direct (2025). DOI: 10.1016/j.preme.2025.100049
Am J Physiol Cell Physiol. (2020). DOI: 10.1152/ajpcell.00120.2020

NEW GUIDELINES FLIP THE FOOD PYRAMID

BY ELLIS ROSENBERG, CELL & MOLECULAR BIOLOGY, 2029

DESIGN BY DIVYA GANDHI, CHEMISTRY, 2029



Nutrition advice has changed so often that many people have stopped believing it. Butter was bad, then acceptable; fat was dangerous, then essential; carbohydrates were harmless, until they weren't. As recommendations continue to evolve, a persistent unease remains: if guidance keeps changing, how can we be sure we are eating the right way now?

In the United States, dietary advice has long been shaped at the federal level. The 1992 food pyramid gave way to the MyPlate model in 2011, reflecting updated priorities in nutrition education. Following the 2024 election and subsequent cabinet appointments, Donald Trump appointed Robert F. Kennedy Jr. to lead the Department of Health and Human Services (HHS), which oversees agencies including the Food and Drug Administration, Centers for Disease Control and Prevention, and National Institutes of Health. In January 2026, the U.S. Department of Agriculture (USDA) and HHS released the 2025–2030 Dietary Guidelines, replacing the MyPlate graphic with a new inverted food pyramid — an update that reflects not only changing scientific evidence, but also the evolving relationship between nutrition science and federal policy.

The traditional food pyramid was introduced in 1992 and reflected the best nutritional science and public health opinions. Developed and promoted by institutions such as the USDA, the wide base was breads, cereals, rice, and pasta, encouraging Americans to eat 6–11 servings per day. Above that sat fruits and vegetables, followed by moderate amounts of protein sources such as meat, fish, eggs, and dairy. At the narrow top were fats, oils, and sweets, which were to be eaten sparingly. The pyramid emphasized calorie balance and nutrient adequacy rather than food processing, metabolic effects, or long-term chronic disease risk, making it a practical but simplified representation of nutrition science as it was understood in the late 20th century.

At the time, nutrition education focused largely on macronutrients, calorie balance, and the prevention of vitamin and mineral deficiencies, rather than on metabolic health or long-term chronic disease. Fat was widely viewed as the primary dietary risk factor for heart disease, while carbohydrates were considered a safe and efficient source of energy. Concepts such as insulin resistance, glycemic load, and the effects of food processing were poorly understood or not widely received.

The new food pyramid reflects how nutrition science has evolved alongside changes in disease patterns, food systems, and research methods. Unlike the traditional model, which was shaped by mid-to-late-20th-century concerns about nutrient deficiencies and calorie balance, the updated pyramid incorporates scientific results from decades of data on metabolism, insulin regulation, and chronic disease risk. As ultra-processed foods became more common and rates of obesity and type 2 diabetes increased, researchers gained clearer evidence that food quality and processing matter as much as macronutrient totals.

In response, the new pyramid shifts its foundation toward vegetables, fruits, legumes, and minimally processed foods, while placing healthy fats and high-quality proteins in a more prominent role. Refined grains, added sugars, and heavily processed foods are now positioned at the top, reflecting their association with metabolic dysfunction rather than essential nutrition. This reorganization aligns with contemporary guidance from agencies such as the USDA and the World Health Organization, illustrating how dietary guidelines adapt as scientific understanding deepens.

Rather than being "inconsistent," the inverted pyramid demonstrates how nutrition recommendations are updated to reflect new evidence and changing realities. The evolution of the food pyramid should not be seen as a failure of science, but as evidence of how nutrition and public health advance as new data emerge.

National Library of Medicine (2025). DOI: 10.1186/s12889-025-22304-3
National Library of Medicine (2018). DOI: 10.1093/advances/nmx025

PHOTO VIA ADOBE STOCK

BONE BREAKING CLARITY

CONNECTION BETWEEN OSTEOPOROSIS AND NUTRITION

BY JOSEPHINE DERMOND, ECONOMICS, 2026

DESIGN BY DIVYA GANDHI, CHEMISTRY, 2029

Broken bones from risky playground hops, hard ball tosses, and bike accidents are all too common for kids — and worrisome for their parents. However, adults should be equally as concerned for their bones as they are for their childrens'. Osteoporosis, a bone disease characterized by reduced bone density and increased risk for breaks and fractures, is on the rise. The disease is becoming particularly more prevalent in older adults and women.

A systematic review published in 2016 by Connie M. Weaver and team identified that lifestyle factors including nutrient intake contribute 20–40% of the variance in bone mass. Nonetheless, diet research remains convoluted and studies, media, and data on nutrition and bone health are inconsistent, leaving people unsure about how to control their health.

Bones are living tissues, just like other parts of our bodies. They are composed of cells following a build-up and break-down life cycle driven by osteoblasts and osteoclasts. Osteoblasts aid development by strengthening existing tissue and growing new bone cells that act as fortifying varnishes or sealants. Osteoclasts are specialized cells for dissolving or breaking down old bone tissue, like a chemical paint stripper. Though osteoblasts and osteoclasts work harmoniously to rid the bones of unhealthy cells and replace them with new ones, according to Weaver, bones' build-up function significantly deteriorates after 20–30 years of age, while the breakdown function remains steady.

Among crazes surrounding collagen, peptides, combinations of calcium and vitamin D, and even prunes, it's hard to weed out the truth of what might improve bone health. An article written by endocrinologist Meryl LeBoff and her team identified that while vitamin D is necessary for calcium absorption, the majority of people have sufficient amounts and should instead focus on increasing calcium consumption. Eating the approximately ~1300 milligrams of calcium recommended to reach peak bone density is extremely challenging, even with supplementation, because not all calcium that is ingested actually gets absorbed.

Articles such as LeBoff's lead many to protein for answers. In Weaver and colleagues' review, they analyzed a study by Vatanparast Hassanali and team that measures the influence of protein intake on bone mass in the presence of calcium. The study found that protein intake explains about 37% of the conditional variance in net bone mineral content gain in 8–21 year olds with medium–high calcium levels. This means that protein's effect on bone strength was not statistically significant but had some influence on bone mineral content. Therefore, while this study does not necessarily imply that increases in protein intake will guarantee improved bone mineral content, it does indicate protein is statistically associated with it for that subgroup.

Similar to the benefits from protein consumption, the association between fat intake and bone density are contested. The results of different fatty acids — a component of the macronutrient fat — showed a range that covers both negative and positive outcomes to bone density depending on the fatty acid. Interestingly, a UC Davis study reports that a maternal brain hormone (CCN3) prevalent in pregnant mice fed high-fat diets stimulated bone formation and strength; therefore, certain demographic characteristics may play a role.

While lifestyle factors such as diet are less effective after age 30, hope for youth remains, considering that 10% increases in bone mass density before the cutoff age can delay the onset of osteoporosis by 13 years. This decreases the likelihood of future fractures and breaks for the majority of the population. Without a concentrated effort to bolster the production of osteoblasts before their production stalls, damage to the "varnish" occurs faster than new "coats" can be applied. Therefore, overall diet and consumption patterns may play a larger role than originally understood in setting the foundation for the body's framework, our bones. Further clarification on the relationship between diet and bone health is needed, but balanced nutrition, particularly at a young age, is a solid first step in preventing osteoporosis.

Osteoporosis International (2024). DOI: 10.1007/s00198-024-07031-6
Nature (2024). DOI: 10.1038/s41586-024-07634-3
Biomedicines (2023). DOI: 10.3390/biomedicines11112982
New England Journal of Medicine (2022). DOI: 10.1056/NEJMoa2202106
Journal of Bone and Mineral Research (2022). DOI: 10.1002/jbmr.5628
Osteoporosis International (2016). DOI: 10.1007/s00198-016-3652-6

PERSONAL ESSAY

Mechanical vs. moral challenges in emergency medicine

BY KARLA KUNTASAL, CELL & MOLECULAR BIOLOGY, 2029

DESIGN BY GIULIA WALKER, DESIGN, 2026

Most people correlate the practice of emergency medicine with saving lives, but what does a "saved life" really constitute?

Emergency responders, including myself, have to go through months of training and obtain numerous certifications and licences just to be able to step foot in an ambulance. We are rigorously taught the mechanics of the human body, and everything it takes to keep a patient stable in the event of an emergency. This is what most people would consider "saving" a life. But what happens when that person arrives at the hospital only to find out they will be bedbound for the rest of their life? Is their "life" really saved? Medical professionals often save a patient's life mechanically, but the "life" of the patient can be severely altered, and in some cases, lost.

During my first year as an emergency medical technician (EMT), I responded to a cardiac arrest call on an elderly woman. Upon arrival, my crew and I confirmed she had no pulse and started compressions. As an advanced life support (ALS) unit, a paramedic engine, and the battalion chief entered the scene, organized chaos began. In an ALS call, EMTs are at the bottom of the food chain and often just a helping hand as the two paramedics take over. This typically results in EMTs holding fluids that are being pushed, bagging the patient or performing compressions as the paramedics administer rounds of epinephrine, starting intraosseous (IO) lines and actively watching over the cardiac rhythm of the patient.

On this call, I was responsible for holding the intravenous fluid which allowed me to mentally take a step back from the active code and observe what was happening around me. The patient's daughter was on the phone just a few feet away from me crying out, "Please help my mom," followed by, "Dad you need to come, I think Mom is passing away." At this moment, my empathy poured out for her. It made me realize that in cases that involve an abundance of emotion, the best strategy for effectively practicing medicine can be to view patients' injuries as mechanical problems, because the second you start to think of them as someone's mother or daughter, empathy takes over and distraction sets in. I could only imagine that her daughter's heart was beating as fast and hard as the beats being artificially administered to her mother.

After administering compressions for almost 30 minutes, two rounds of epinephrine to promote circulation, and multiple pulse checks that resulted in an asystole (flatline) rhythm, it was time to call time of death. A family night of relaxing on the couch had turned into the unimaginable. Just like that, not only was the patient's life lost, but the life of her family would never be the same again.

The body of a patient that does not survive a cardiac arrest is usually left at the scene and care is transferred to law enforcement. Emergency Medical Services (EMS) providers are expected to get back on the ambulance and start a fresh call. We aren't taught how to deal with this in EMT school. It is the strangest feeling, walking out of a scene we had just arrived at, only to leave behind a distraught family and a deceased individual. It is truly an utterly heartbreaking mess. Most of the emergency personnel on scene will pat each other on the back before leaving and congratulate each other on the smoothness of an intubation or the speed of putting on the LUCAS device. This is because we are taught to take each call as an opportunity for learning and growth. The chance to perform CPR on a real person instead of a dummy comes as an excitement, but is then juxtaposed by the guilt of feeling this emotion. With death comes the destruction of not only the patient's life, but the people who made their life worth living.

A lovely homeless man, let's call him

Teddy, was another patient who caused me to confront this issue. We had once again responded to a cardiac arrest which ended in a "success." We had effectively gotten a pulse back. From a mechanical and textbook standpoint, the call was a success. We had done our jobs. But there was one thing I couldn't get off my mind. Teddy's life had been saved only to be discharged back to living on the street. His "saved" life still consisted of great vulnerability and financial hardship.

> **"The challenge of emergency medicine goes far beyond protocols and textbook definitions; it is a matter of learning to work in between mechanical success and moral uncertainty."**

So was his life really saved? Or was it simply prolonged?

In emergency medicine, every second is critical. Our training is built around airway, breathing and circulation — measurable indicators of life. We are taught to restart hearts and force oxygen into lungs in order to keep a patient stable. In these moments, patients must become physiological problems rather than people with histories, relationships, and futures. If we allow ourselves the emotional magnitude of the scene, distraction could cost a life.

Once the paperwork is filed, and the ambulance is cleaned and prepped for the next call, the moral questions still linger. The woman whose family watched her final moments. The homeless man who survived only to return to hardship. These experiences reveal the quiet truth of emergency medicine: that we can often save the body of a patient, but we cannot control what that saved body returns to. We treat the mechanics of survival, not the circumstances that define its quality.

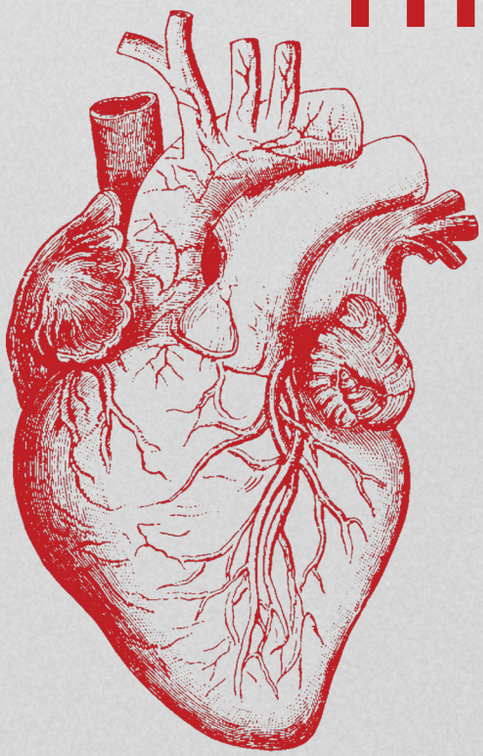
The challenge of emergency medicine goes far beyond protocols and textbook definitions; it is a matter of learning to work in between mechanical success and moral uncertainty. It is performing compressions with precision while knowing that survival does not always mean restoration. It is celebrating the return of spontaneous circulation while questioning what type of life that pulse sustains.

In the end, emergency medicine does save lives. But it ultimately forces providers to confront a deeper question: what does it truly mean to save a life? Is it the return of a heartbeat, or the preservation of a life worth living? The answer is rarely clear. And this uncertainty, more than the blood, late nights, and chaos, is the most difficult challenge of all.

PHOTO VIA ADOBE STOCK

THE HEART OF THE PROBLEM

Do beta-blockers really help all heart attack patients?



BY GABRIELLE WEINER, BIOLOGY, 2026
DESIGN BY ANJANA BALAKRISHNAN, STUDIO ART, 2027

With each heartbeat, the left ventricle propels oxygen-rich blood into the aorta to be distributed to nearly every tissue in the body. The efficiency of this chamber of the heart is routinely assessed in clinical settings, with a measurement called *left ventricular ejection fraction* (LVEF), which assesses the damage after a heart attack. Since the 1980s, the standard of care after myocardial infarction (a heart attack) has been to administer beta-blockers — medications which soften the effects of adrenaline, slowing heart rate and relaxing blood vessels. Even patients with a preserved LVEF (50% or greater) receive them. Since the development of modern myocardial infarction treatments, mortality rates have declined dramatically, indicating that this approach is effective. But for some scientists, *effective* doesn't necessarily mean *optimal*. Earlier this year, a team of researchers published a study in the *New England Journal of Medicine* that challenges the routine use of beta-blockers, suggesting that they might not be necessary for all patients.

The team conducted a meta-analysis of five clinical trials that assigned heart attack patients with preserved LVEF to either receive or not receive beta-blocker therapy. In total, 17,801 patients were included: 8,831 (49.6%) of those patients were assigned to receive beta blockers, while 8,970 (50.4%) were in the no-beta-blocker group. In the time following their heart attack, 335 patients in the beta-blocker group died, compared to 326 patients in the no-beta-blocker group. Of those deaths, those that were cardiac-related occurred at similar rates — nearly the same between the beta-blocker and no-beta-blocker groups. To no surprise, occurrences of heart failure followed suit, happening in about the same number of patients from both groups. After accounting for factors such as age, sex, and cardiovascular history, each comparison resulted in that same conclusion: patients with preserved LVEF and no other indication for the medication seem to gain no benefit from beta-blocker therapy.

Of course, no benefit doesn't necessarily mean that beta-blockers hurt the prognosis of patients with preserved LVEF. What's the incentive to change medical tradition? While beta-blockers save lives, they aren't completely harmless. Patients are at risk of a variety of side effects, like exercise intolerance and hypotension. Aside from a better quality of life, individuals with preserved LVEF also stand to experience a lower medication and financial burden.

Current treatment for myocardial infarction has been reinforced through decades of research and clinical application. A handful of studies with a population of 17,000 people isn't enough to change the standard of care — in the US alone, over 800,000 people experience a heart attack each year. Even among patients with preserved LVEF, residual ischemia or a high resting heart rate could persist, things that could easily respond to beta-blockers. Still, the meta-analysis stands to upend long-standing clinical assumptions, something that's incredibly hard to do in medicine. It also highlights the power of the left ventricle: small changes in its function can result in completely different clinical outcomes.

New England Journal of Medicine (2026) DOI: 10.1056/NEJMoa2512686

THE MIND-HEART CONNECTION

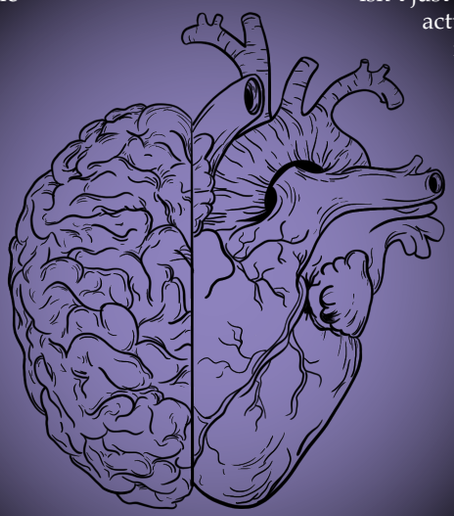
Depression's role in cardiovascular health

BY CECELIA KINCAID, SPEECH-LANGUAGE PATHOLOGY & AUDIOLOGY, 2027

DESIGN BY DIVYA GANDHI, CHEMISTRY, 2029

Depression is usually conceptualized as a disorder of the brain, but growing evidence suggests that it is tied to heart disease. Psychological stress may eventually manifest as cardiovascular risk.

Research as early as 2002 has found significant evidence that depression can serve as a predictor for coronary heart disease (CHD), which is a leading cause of death worldwide. Efforts to identify psychological predictors for CHD began in the 1950s, but studies initially yielded inconsistent results. With a massive increase in evidence after the turn of the century, researchers controlling for traditional risk factors found that people with depression have higher rates of a CHD diagnosis or cardiac events later in life. This suggests that depression is not only comorbid with CHD, but may be a key clinical predictor.



A 2012 review corroborated this finding and established a more explicit relationship. Depression increases the risk and incidence of coronary artery disease, and also influences the outcomes of those who already have heart disease, usually correlating with a worse prognosis. Those with depression tend to have higher mortality rates after a heart attack, heart failure, or even common heart surgeries such as mitral valve replacement. The idea that severity of depression can influence prognoses and the risk of adverse cardiac events was also illustrated in a 2024 review. The link between depression and cardiovascular health is increasingly clear. But how exactly does a psychological disorder increase cardiovascular risk?

There is no single explanation for how depression plays a role in heart health. One of the more important mechanisms is inflammation. People with depression have elevated pro-inflammatory cytokines, or signaling proteins, which can destabilize heart muscle by increasing macrophage action and triggering cell death. The severity of depression is correlated with the increase of cytokines.

Another mechanism is dysregulation of the HPA axis, which controls the body's stress response. Depression is associated with chronic activation of the HPA axis, which raises cortisol levels. High cortisol can severely increase cardiovascular risk, mainly by affecting heart rate, blood pressure, fat accumulation, and insulin resistance.

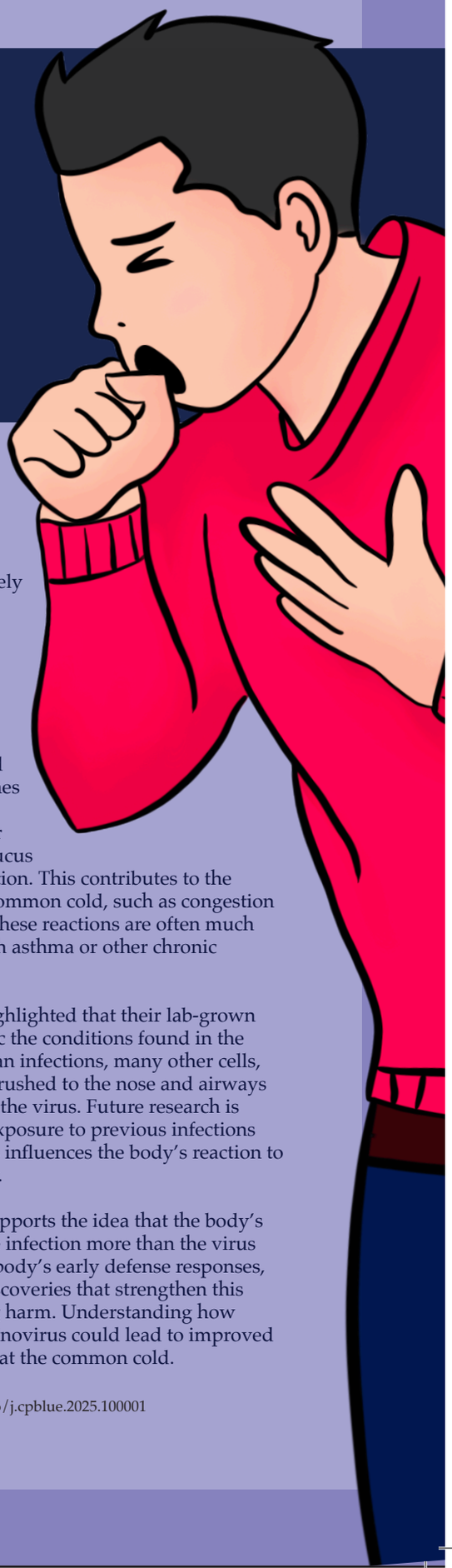
A third important mechanism is autonomic dysfunction. Depression leads to overactivation of the sympathetic nervous system, known for the fight-or-flight response, and underactivation of the parasympathetic nervous system, responsible for maintaining homeostatic vitals and conserving energy. Overactivation of the fight-or-flight system, which involves increased heart rate, blood pressure, and demand for oxygen, can cause coronary artery spasms or constriction of the blood vessels.

There are quite a few biological pathways that are affected by depression beyond these three. Depression isn't just correlated with heart health; it can actually contribute to pathophysiology in cardiovascular disease, even in the absence of traditional risk factors like smoking. Additionally, it's crucial to remember that these systems influence each other in a loop. Psychological factors influence biological pathways, which can in turn worsen psychological health. Essentially, depression can contribute to pathophysiology and therefore increase depressive systems.

The question is, how can the complex relationship between biology, psychology, and cardiology help inform better treatments for cardiovascular patients? Since depression can influence the prognosis, symptom severity, and quality of life in heart disease patients, clinicians need to consider the neural-cardiac link to create effective treatments and preventative care strategies. Many researchers emphasize the importance of screening for depression to identify cardiovascular risk. Others advocate for holistic care, rather than treating depression and the heart separately in the individuals with both health concerns. One possible avenue is to use antidepressants as a treatment in comorbid CHD patients, but these studies yield mixed results. The best solution will likely involve a combination of pharmacological and non-pharmacological approaches. In the meantime, clinicians can use the strong interplay between the mind and the heart as a guide when treating patients with depression, cardiovascular disease, or both. Future research will hopefully identify the key to integrated healthcare by learning more about the way the mind and the heart influence one another.

Frontiers in Psychiatry (2024). DOI: 10.3389/fpsyt.2024.1328048
Current Problems in Cardiology (2023). DOI: 10.1016/j.cpcardiol.2023.101987
Nature Reviews Cardiology (2012). DOI: 10.1038/nrcardio.2012.91
American Journal of Preventative Medicine (2002). DOI: 10.1016/S0749-3797(02)00439-7

WHY COLDS AFFECT PEOPLE DIFFERENTLY



BY MARINA IANNACITO, BIOLOGY, 2028
DESIGN BY DIVYA GANDHI, CHEMISTRY, 2029

Why do colds leave some people feeling awful, while others barely feel sick? New research indicates that this difference may come from the nose and its early immune defenses, which can stop a virus before symptoms begin. The common cold is primarily caused by rhinovirus, a contagious infection that affects the upper respiratory tract, including the linings of the nose, throat, sinuses, and trachea. But scientists believe the virus itself does not determine the severity of the cold. Instead, the intensity is shaped by the speed and strength of your body's initial response.

> **"Instead, the intensity is shaped by the speed and strength of your body's initial response."**

A research team at Yale School of Medicine began investigating this theory by culturing lab-grown human tissue using nasal stem cells. For four weeks, the tissue was exposed to air to promote maturity into a structure that strongly mirrors the lining of the nose and lungs, with mucus-producing cells and cells with cilia, tiny hairs that help transport mucus and other particles out of the respiratory tract. Using these creations, the scientists could see how thousands of human nasal cells respond to rhinovirus. As rhinovirus does not infect any animals other than humans, making this model was key for studying the virus and its replication.

Upon detection of rhinovirus, the cell releases interferons, which activate antiviral defense in both infected cells and nearby healthy cells, making it more difficult for the virus to replicate in the body. If the interferons reacted quickly, the virus struggled to multiply, and the infection was contained. On the other hand, when the scientists intentionally blocked interferons from being produced, the virus spread extremely quickly, greatly damaging many cells and even destroying the tissue entirely in some cases. The timing of the nose's defense system largely dictates how intense the infection becomes before the immune system attacks.

The researchers also found that when the virus becomes more widespread, it can activate additional cellular pathways that increase mucus production and inflammation. This contributes to the typical symptoms of the common cold, such as congestion and difficulty breathing. These reactions are often much more severe in people with asthma or other chronic respiratory conditions.

However, the scientists highlighted that their lab-grown tissue does not fully mimic the conditions found in the human body. In true human infections, many other cells, such as immune cells, are rushed to the nose and airways to help limit the spread of the virus. Future research is imperative to learn how exposure to previous infections and environmental factors influences the body's reaction to rhinovirus and its severity.

Nonetheless, this study supports the idea that the body's reaction may influence the infection more than the virus itself. By looking into the body's early defense responses, scientists aspire to find discoveries that strengthen this reaction while minimizing harm. Understanding how nasal cells engage with rhinovirus could lead to improved methods to reduce and treat the common cold.

Cell Press Blue (2026). DOI: 10.1016/j.cpblue.2025.100001

IMAGE VIA ADOBE STOCK

PREDICTING THE UNPREDICTABLE

How AI is transforming epilepsy care

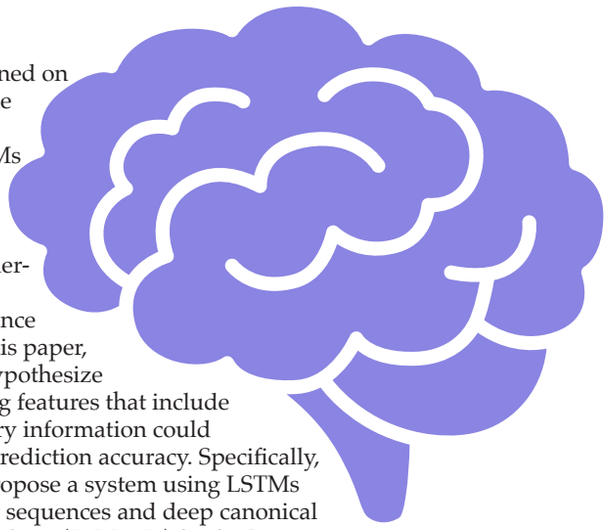
BY MATTHEW GARCIA, ELECTRICAL & COMPUTER ENGINEERING, 2026
DESIGN BY CANDICE MEDINA, MECHANICAL ENGINEERING, 2026

Epilepsy is the third most common neurological disorder in the United States, affecting about 3.4 million people — 1% of the population. One of the biggest challenges articulated by the epilepsy community is the unpredictability of seizures. While epileptic seizures do follow patterns that can broadly be described, pinpointing when exactly a seizure will occur in the moments before is a more challenging issue. Despite this, there is growing evidence that human biomarkers may show warning signs before a seizure occurs.

One example of this is seizure dogs, whose increased olfactory capabilities are able to be used to distinguish between seizure and non-seizure sweat, most likely due to the presence of volatile organic compounds (VOCs). A 2024 paper suggests that dogs were able to predict seizures from sweat samples in 94% of epileptic events. These findings indicate that there may be a measurable biological change that has strong predictive capability for seizures. In this sense, artificial intelligence methods can be employed in order to learn the features of these biomarkers for prediction.

Recent research also examines the most relevant models needed to predict seizures from EEG signals. After conducting a literature review, authors identified a study employing a deep learning model, known as long short-term memory (LSTM), that can detect seizure patterns in brain wave data within 15-120 minutes before a seizure occurs with a sensitivity of 99%. The study found LSTM models achieved an average accuracy of 93%, with average warning time of 21 minutes. Using another deep learning model known as a transformer, an additional study referenced by the authors was able to achieve a prediction sensitivity of 96%, with a warning time of 3 to 30 minutes. This high performance originates from the unique ability of these models to work with time-series data. For LSTMs, this is due to the model's ability to process sequences iteratively while capturing long-term dependencies. For transformers, this is due to the direct connections between different tokens within the sequence. While these results are promising, there may be difficulty in using EEG data in the real world, as these procedures are considered invasive.

Using non-invasive methods, a 2025 research paper showed that autonomic nervous system (ANS) biomarkers, such as heart rate and temperature, can also be useful for predicting changes shortly before the seizure. Prior methods for predicting seizures using biomarkers obtained through wearable devices include supervised learning models, which are trained on using wearable data as input features. LSTMs are also used here, being trained on raw and Fourier-transformed data for sequence learning. In this paper, the authors hypothesize that combining features that include complementary information could improve the prediction accuracy. Specifically, the authors propose a system using LSTMs for processing sequences and deep canonical correlation analysis (DCCAEs) for finding correlations between ANS subsystems. This system was used in conjunction with a growing neural network, which allowed for the system to integrate different features from these regions as the architecture of the machine learning model was changed. Through this, they found that combining multiple biological signals significantly improved prediction accuracy, raising it from about 74% to 82%. This suggests that seizures may affect several body systems at once, and models that integrate multiple data streams perform better.



Looking ahead, researchers hope to develop wearable systems that employ transfer learning and growing neural networks to continuously learn and adapt to each individual's seizure patterns. By combining brain activity, heart rate, and temperature, future devices may provide earlier and more reliable warnings. Returning to the past idea of VOCs, this hybrid of generalizable models and wearable methods can play a key role in creating a new, non-invasive input stream for seizure prediction. If successful, these technologies could transform seizure prediction from a clinical tool into an everyday safety system.

bioRxiv Preprint (2025). DOI: 10.1101/2025.01.28.635212
Epilepsy & Behavior (2024). PMID: 38071830
Nature Reviews Neurology (2024). DOI: 10.1038/s41582-024-00965-9
Journal of Integrative Bioinformatics (2023). PMID: 38099461
Scientific Reports (2022). PMCID: PMC9243143
Applied Sciences (2022). DOI: 10.3390/app12094158
Computers in Biology and Medicine (2018). PMID: 29807250

PHOTO VIA ADOBE STOCK

Caffeine for cognition

BY AVERY KIM, BIOENGINEERING & BIOCHEMISTRY, 2028
DESIGN BY CECELIA CLEMENTE, CHEMICAL ENGINEERING, 2028

Latte, americano, cappuccino, matcha, cold brew, french press — a multitude of options exist for those wanting that jolt of caffeine in the morning — and for good reason, seeing as the average American consumes roughly three cups of coffee a day. But caffeine may have benefits beyond providing that morning boost of energy.

A recent study by a team of researchers from the Harvard T.H. Chan School of Public Health and the Brigham Women's Hospital showed that moderate daily caffeine consumption could potentially be linked to a decreased risk of dementia and less cognitive decline with age. Data collected from the Nurse's Health Study (NHS) and the Health Professionals follow up study (HPFS) provided the team with information on 131,821 male and female health care professionals over 43 years. While previous studies have investigated the benefits of caffeine for cognitive health, none were performed to this extent or provided data longitudinally.

> **"Optimal results for both decreased dementia risk and increased cognitive function were for those who consumed roughly 300 mg of caffeine daily, which is equivalent to two to three cups of coffee."**

The team tracked the amount of caffeine consumed by the individuals over 43 years, and had the participants complete questionnaires to test for cognitive function. Caffeine intake was included for a variety of beverages including coffee, tea, and soda, as well as in certain foods such as chocolate. However, the main source of caffeine came from caffeinated and decaffeinated coffee as well as from tea.

The participants were then divided into four quartiles based on their caffeine intake, with quartile one being those with the lowest caffeine intake and quartile four being those with the highest caffeine intake. The primary finding of the study was that those who consumed more caffeinated coffee, quartile four, demonstrated a lower risk for dementia, with the dementia incidence rate being only 141 out of 100,000 individuals, compared to those in the quartile one, where the dementia incidence rate was 330 out of 100,000.

The results followed the same trend for high tea intake, though curiously did not replicate for decaffeinated coffee consumption.

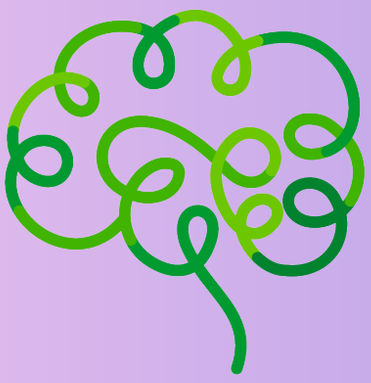
In addition to the findings on dementia, it was also revealed that caffeinated coffee and tea were predictors of decreased cognitive decline. Specifically for the individuals in the NHS, caffeinated coffee consumption was linked to higher cognitive function based on the questionnaires. Optimal results for both decreased dementia risk and increased cognitive function were found for those who consumed roughly 300 mg of caffeine daily, which is equivalent to two to three cups of coffee.

While the study presents only a correlational relationship between caffeine consumption and improved cognitive function and dementia outlooks, caffeine has been established to have positive effects on short and long-term memory formation, alongside its more well-known effects of improving alertness and increasing focus.

With the amount of caffeine the average adult consumes, there is more research to be done on the longer-term effects of caffeine that go beyond providing that morning pick-me-up. So, the next time you reach for your caffeinated beverage of choice, you can keep in mind that you may not only be helping yourself stay awake for the day, but also helping your brain stay sharp for years to come.

Journal of the American Medical Association (2026). DOI: 10.1001/jama.2025.27259
Cureus (2021). DOI: 10.7759/cureus.15032

PHOTO VIA ADOBE STOCK



High on the mind

The effects of cannabis on cognitive functions

BY NINA ZHENG, BEHAVIORAL NEUROSCIENCE, 2029

DESIGN BY CECELIA CLEMENTE, CHEMICAL ENGINEERING, 2028

For the past decade, traditionalism has taken a progressive turn. Things that were considered "taboo" were legalized to fit current social and economic trends. One of the more controversial legalizations is the accessibility of certain drugs, including marijuana, due to its medical and recreational usages. Cannabis, now legal across 21 states, proves a bigger impact on society as millions of Americans wish to seek its promised effects. A big portion of this population includes young adults, many of whom report having used weed at least once before they even finish high school. Concerningly, cannabis use during adolescence has been linked to long-term effects on impulse control and major executive functions, which prompts the countrywide legal age to purchase cannabis to be at least 21 years of age. However, this doesn't stop the cannabis revolution, as thousands of dispensaries, or even illegal vendors who sell to those that don't meet the legal criteria, emerge in hopes of satiating its popularity. With THC concentrations presumably higher than in previous decades, worries persist about whether repeated usage can lead to subsequent negative consequences on cognition.

Despite the negative light often associated with marijuana, there are some benefits of the drug. In medicine, cannabis can be an effective pain reliever, easing diseases ranging from Parkinson's to general nerve pains. Medicinal cannabis is one of the leading reasons why people pushed to legalize this drug, as it can mimic the effects of the body's naturally occurring "happy" hormones. However, even though marijuana can provide almost instantaneous relief, patients should still treat it with caution.

As weed becomes increasingly popular, researchers are trying to evaluate the correlation between cannabis and long term cognitive changes. Amidst the developing studies, one thing remains clear: adolescence is a critical period for brain development. Young adults, especially teenagers, are one of the main consumers of this substance. In the brain, a signaling system called the endocannabinoid is responsible for stress regulation, motivated behavior, development, and the sculpting of neural circuits, all of which are processes that continue through adolescence.

The key players in the endocannabinoid system are molecules called anandamide and 2-AG (2-arachidonoylglycerol), which are primary activators that bind to cannabinoid receptors in the brain called type 1 (CB1) and type 2 (CB2). THC, the main psychoactive found in many recreational and medical marijuana products, also binds to the CB1 receptor, which in turn activates the endocannabinoid system. Unlike anandamide and 2-AG, which are naturally produced by the body, consistent use of THC can overtax the CB1 receptors, disrupting the needed developmental signaling pathways to occur.

Use of cannabis may be first driven by curiosity, or simply a lack of awareness. Most people think that a marijuana overdose isn't possible as it is with other hard drugs. This is partially true, as marijuana doesn't impact the body the way fentanyl and other opioids do, but ingesting too much can lead to extreme intoxication. This phenomenon, known as "greening out," is characterized by a variety of symptoms including panic, palpitations, and nonstop vomiting. Coming back from the high can also lead to an array of adverse effects, such as brain fog, which can impact crucial critical thinking and quality of life.

The misperceptions of marijuana have been debated for years. Many hope it will address health issues or enhance their well-being if other options have fallen short. Understanding the effects of marijuana is crucial as the cannabis market seems to boom year after year. In a society where stress, pressure, and burnouts are increasingly common, individuals turn to drugs as a form of relief. People celebrate it. Businesses' profits grow. Celebrities endorse it and some even make it their brand. Figures such as Snoop Dogg have openly embraced cannabis culture, bridging a connection between marijuana and pop culture. Yet, as the stigma around cannabis fades, a question remains unanswered: is society really acknowledging the neurological costs of momentary leisure?

JAMA Network Open (2025). DOI: 10.1001/jamanetworkopen.2024.57069

PHOTOS VIA ADOBE STOCK

COGNITIVE DISSONANCE

How choices are shaped

BY AYAH KURDI, BEHAVIORAL NEUROSCIENCE, 2029

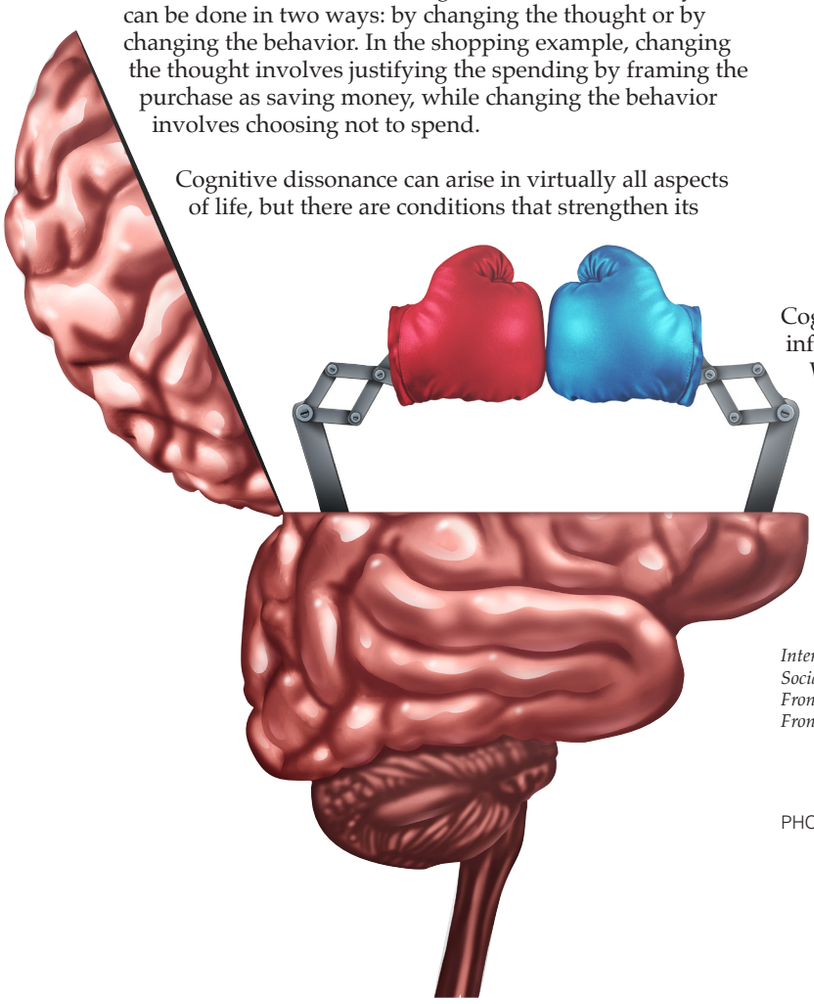
DESIGN BY CANDICE MEDINA, MECHANICAL ENGINEERING, 2026

Black Friday arrives with irresistible deals and urgent messages: "once a year" and "too good to miss." Carts fill with items never intended to be bought, justified by the thought that the money would be spent eventually, so the savings must be worth it. Crowds of people drown out the question of whether any of it is truly needed.

Discomfort can arise when one holds two contradictory beliefs; this is the essence of cognitive dissonance. For example, the two thoughts of saving and spending are inconsistent, creating psychological tension. This theory was first presented by American social psychologist Leon Festinge in 1957 in his book, "A Theory of Cognitive Dissonance." Building on his social comparison theory, which proposed that people have an innate drive to evaluate one's attitude compared to others, he questioned how individuals navigate discrepancies between their own actions and beliefs.

When dissonance occurs, the mental discomfort must be reduced to maintain mental regulation and consistency. This can be done in two ways: by changing the thought or by changing the behavior. In the shopping example, changing the thought involves justifying the spending by framing the purchase as saving money, while changing the behavior involves choosing not to spend.

Cognitive dissonance can arise in virtually all aspects of life, but there are conditions that strengthen its effect. When an action is freely chosen, there's room for internal conflict, and if it's also under high commitment, the choice can feel more meaningful. The dissonance is also greater if the action causes negative consequences, such as financial distress, especially when those outcomes were foreseeable at the time of the decision. These conditions shape how people reconcile their attitudes and behaviors.



Beyond shaping individual beliefs and actions, cognitive dissonance is a fundamental mechanism for evolution. New knowledge is only useful if it modifies what is already known, often conflicting with existing beliefs, which creates cognitive dissonance. When new information challenges what's established, society reduces this tension through either rejection or rationalization. Over time, new knowledge proves its value and becomes integrated into what's known, shaping human learning and reasoning.

> **"While it typically influences behavior unconsciously, understanding and being actively aware of cognitive dissonance can be an opportunity for purposeful, positive change."**

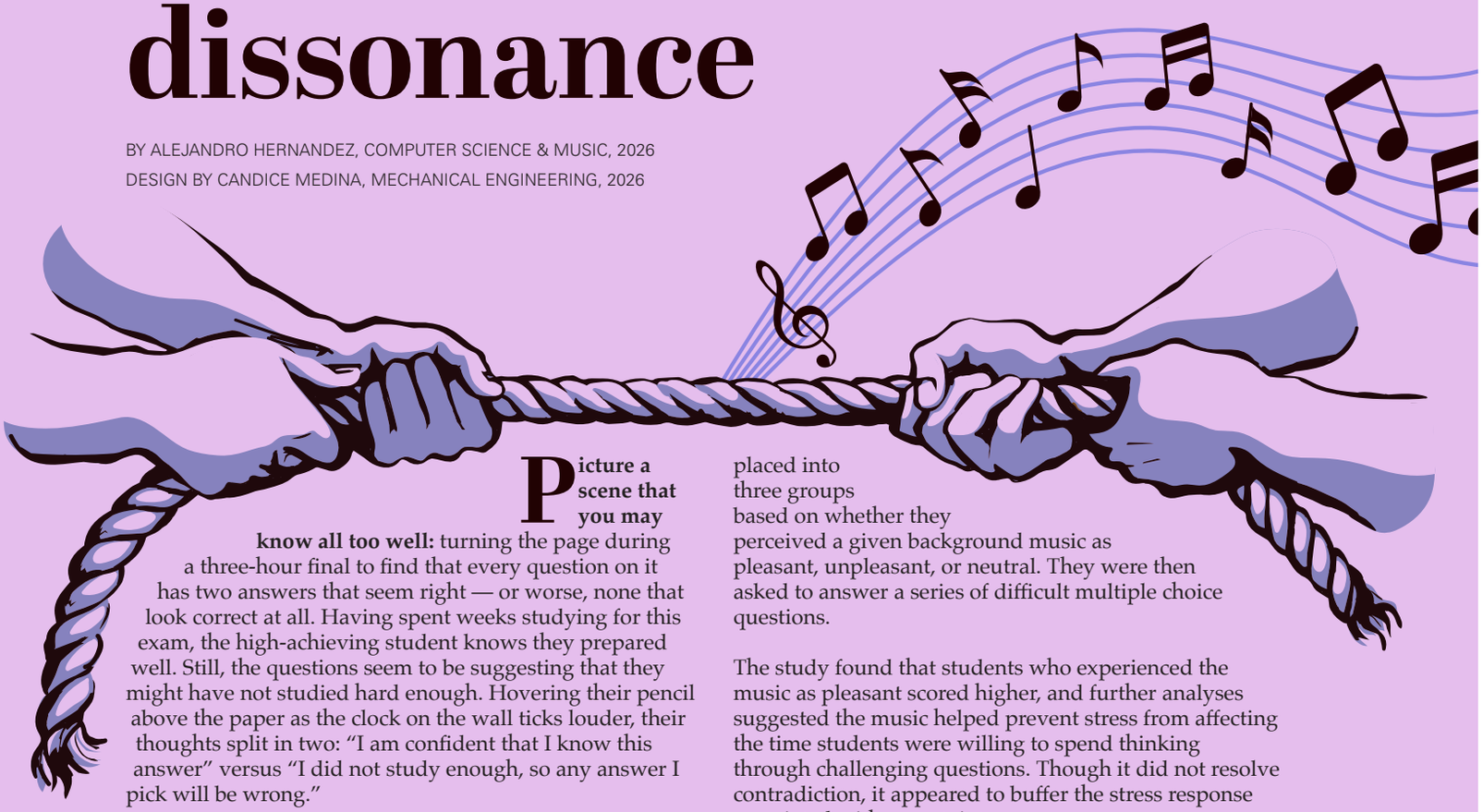
Cognitive dissonance is considered one of the most influential and studied theories within social psychology. When in a state of dissonance, the process of reduction directly influences individual belief perceptions and actions, serving as an underlying propeller of cognitive evolution. While it typically influences behavior unconsciously, understanding and being actively aware of cognitive dissonance can be an opportunity for purposeful, positive change.

International Review of Social Psychology (2019). DOI: 10.5334/irsp.277
Social and Personality Psychology Compass (2017). DOI: 10.1111%2Fspc3.12362
Frontiers in Psychology (2013). DOI: 10.3389/fpsyg.2013.00179
Frontiers in Psychology (2010). DOI: 10.1093/erae/jbq012

PHOTO VIA ADOBE STOCK

Mozart's effect on dissonance

BY ALEJANDRO HERNANDEZ, COMPUTER SCIENCE & MUSIC, 2026
DESIGN BY CANDICE MEDINA, MECHANICAL ENGINEERING, 2026



Picture a scene that you may know all too well: turning the page during a three-hour final to find that every question on it has two answers that seem right — or worse, none that look correct at all. Having spent weeks studying for this exam, the high-achieving student knows they prepared well. Still, the questions seem to be suggesting that they might have not studied hard enough. Hovering their pencil above the paper as the clock on the wall ticks louder, their thoughts split in two: "I am confident that I know this answer" versus "I did not study enough, so any answer I pick will be wrong."

That mental tug-of-war is known as cognitive dissonance, the psychological tension of holding two contradictory beliefs simultaneously. During these moments, the brain becomes defensive and scrambles for relief by rushing to an answer to escape the discomfort, second-guessing decisions, or just by shutting down. As it turns out, music may be able to help. In the early 1990s, psychologists began debating the idea that listening to classical music could temporarily boost cognitive performance on spatial reasoning tasks. This claim soon gave way to a phenomenon known as the "Mozart effect." However, the nuance quickly disappeared once the public began to see it as proof that classical music simply makes you smarter. Within a few years, the effect was re-framed as modest, short-lived, and possibly unrelated to music altogether.

By 2013, researchers returned to this debate with a different theoretical framework to ask which cognitive processes might be changing during improved performance. When tension becomes intolerable, individuals may slow their learning processes by avoiding or suppressing conflicting ideas. But if learning depends on confronting contradiction, then is the ability to tolerate dissonance central to intellectual development? This led researchers toward a new hypothesis: can music help students push beyond the stress of test-taking anxiety? To test it, students were placed into three groups based on whether they perceived a given background music as pleasant, unpleasant, or neutral. They were then asked to answer a series of difficult multiple choice questions.

The study found that students who experienced the music as pleasant scored higher, and further analyses suggested the music helped prevent stress from affecting the time students were willing to spend thinking through challenging questions. Though it did not resolve contradiction, it appeared to buffer the stress response associated with uncertainty.

> **"That mental tug-of-war is known as cognitive dissonance, the psychological tension of holding two contradictory beliefs simultaneously."**

Music has been a prominent force in many cultures throughout history, and upon close examination, its persistence seems less accidental and more adaptational. If pleasant music can increase our tolerance for uncertainty, it may be one of the oldest cognitive support systems we carry, especially when pressure is high. So, the next time you turn the page and find your thoughts splitting in two, remember: it might not be about intelligence at all.

Behavioral Brain Research (2013). DOI: 10.1016/j.bbr.2013.01.036
Nature (1993). DOI: 10.1038/365611a0

PHOTOS VIA ADOBE STOCK

HOW DO YOU MEASURE CREATIVITY?

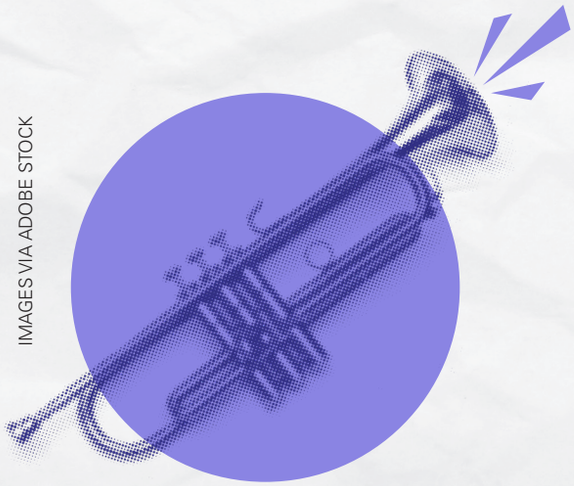
BY KAASHYAP BALAJI, BEHAVIORAL NEUROSCIENCE, 2028
DESIGN BY ANJANA BALAKRISHNAN, STUDIO ART, 2027

While there are many ways it can be implemented, creativity serves to enhance and innovate upon the established norm. In a way, every invention, composition, or scientific breakthrough can be traced back to a single point of inspiration, or a single moment of desire to push past a commonly accepted boundary and venture into unknown territory. Yet, for all its influence, we largely don't understand exactly how creativity has changed us.

The process of quantifying creativity is still arbitrary and heavily influenced by subjectivity. To this end, researchers at the Center for Music in the Brain at Aarhus University in Denmark have recently applied objectivity to rating musical creativity. Additionally, improvisation, a form of expressed creativity in music, is common across many disciplines worldwide: Baroque, Chinese Traditional, Griot, Carnatic, and Jazz, to name a few. The researchers focused on Jazz music due to its implementation of structured creativity, which, as opposed to free creativity, fits within the bounds of the musical style while allowing for individual improvisations.

The research team used both resting-state and task-based functional MRI (fMRI) to better understand how creativity physically affects the brain in real time. After asking participants to perform musical tasks with improvisational elements, the researchers found that the brain doesn't have a binary on-off switch for creativity. Instead, it continuously shifts which regions of the brain and networks are most active, depending on how much freedom the musician gives themselves.

When improvisation was structured and melody-guided, the brain relied on regions involved in deliberate decision-making and monitoring those changes. In contrast, when musicians were given more freedom to play spontaneously, activity surged in areas tied to hearing, movement, and pleasure. At the highest levels of creative freedom, the brain cast an even wider net of regions — including those involved in language, complex behavior, and motor control — all firing in connected harmony. Additionally, prior research has confirmed neuroplastic changes in both anatomical structure and functional connectivity as a direct result of musical performance. In short, the brain physically reshapes itself around the creative demands placed upon it. The Center for Music in the Brain's discoveries reach far beyond the jazz discipline, and perhaps even music. Creativity, long treated as an intangible and individualistic pursuit (and therefore separate from the more rigid sciences), leaves a measurable imprint on the brain's architecture. The act of improvising and pushing past the established norm produces structural and functional change akin to training a muscle to be better prepared for its own use. In this sense, anyone who incorporates creativity through improvisation can rewire themselves to be better suited for future creative endeavors.



IMAGES VIA ADOBE STOCK

Neuroscience News (2025). www.neurosciencenews.com/music-improvisation-neuroplasticity-30061/
Aarhus University Pure Portal (2025). DOI: 10.1111/nyas.70042
Asia Technology Research Institute (2025). DOI: 10.71113/JMSS.v2i1.115
Scientific Reports (2021). DOI: 10.1038/s41598-021-98332-x

New research links schizophrenia to auditory hallucinations

BY MADISON CALVANESE, CELL & MOLECULAR BIOLOGY, 2029

DESIGN BY GIULIA WALKER, DESIGN, 2026

Around 80% of people with schizophrenia experience auditory hallucinations. New research from the University of New South Wales suggests that the link between schizophrenia and auditory hallucinations could stem from a schizophrenic brain's misinterpretation of internal dialogue for external speech. Schizophrenia is a chronic psychiatric illness characterized by symptoms such as delusions, hallucinations, disorganized speech, behavior, and cognitive impairment. Interestingly, there is no chemical or biological test for schizophrenia; it is usually only diagnosed after any other possible conditions have been excluded. However, this new study believes that they have identified certain biological markers to look for when diagnosing schizophrenia, which streamlines the diagnosis and treatment process for this disease.

> **"...auditory hallucinations could stem from schizophrenic brains misinterpreting their own internal dialogue for external speech."**

The theory that auditory hallucinations stem from an inner voice in a person's head being misinterpreted as external speech has been suggested but left unproven for many years. Inner speech is what many individuals interpret as "the voice in my head." It narrates thoughts, feelings, and actions, often occurring without the person noticing. Research shows that when a person speaks in their own head or out loud, the part of the brain that processes "outside" noises becomes less active, anticipating the sound of their own voice. "Outside" noises are anything that has been assessed by our brains as sound. For example, when your phone rings, your brain processes that information as "outside" noise. In contrast, a schizophrenia-impeded brain will react as if the inner voice is coming from someone else.

This phenomenon has been studied using an electroencephalogram (EEG), a machine that measures electrical brain activity in the cerebral cortex. Electrical signals in the cerebral cortex are closest to the outer portion of the brain and synchronized, allowing for clear detection by the EEG. When a healthy person speaks out loud or engages in inner dialogue, the electrical activity in their head is decreased. Consequently, signals in the auditory cortex are suppressed, which is crucial for processing sound, alongside helping distinguish an individual's speech from external noise.

When a schizophrenic individual experiences inner dialogue, something interesting happens. Studies have shown that the electrical activity in their brains increases, which creates a dysfunction in signaling between the auditory cortex and other parts of the brain. As a result, the brain interprets the inner voice as coming from someone else. This process explains why auditory hallucinations feel so deceivingly real to patients, despite the speech stemming from their own minds.

The connection between inner dialogue, dysfunction in brain signals, and subsequent auditory hallucinations may be beneficial in the early diagnosis and treatment of schizophrenia. If these symptoms in electrical brain activity can be determined before the onset of auditory hallucinations, patients could receive early treatment. Additionally, this test could lead to a better understanding of the biological causes and symptoms of schizophrenia, leading to more specific and specialized treatments.

Scientific American Magazine (2026) DOI:10.1038/scientificamerican022026-4yxzMCP4IrdujSLJ11AadC
The Journal of Lifelong Learning in Psychiatry (2016) DOI: 10.1176/appi.focus.20160006

PHOTO VIA ADOBE STOCK

Let's have a tea party!

With apes...?

BY AOIFE JEFFRIES, PHARMACY, 2029

DESIGN BY CANDICE MEDINA, MECHANICAL ENGINEERING, 2026

Previously featured in *The New York Times* and on *The Oprah Winfrey Show*, celebrity bonobo Kanzi has taken the spotlight once again for his advanced and almost human-like cognitive abilities. Famous for understanding spoken English, Kanzi has been a popular participant in behavioral research since the early 1990s. Recently, he demonstrated a surprising capacity to partake in pretend play, indicating that apes are capable of surprisingly complex mental processes and raising questions about what truly differentiates the human species from our primate counterparts.

In a study at Johns Hopkins University, researchers staged a sort of "tea party" to test Kanzi's pretend play tendencies. Scientists spoke to Kanzi in English, initiating game play and providing verbal prompts throughout. Once the game began, the experimenter set two cups on the table before Kanzi and pretended to pour into each cup from an empty, transparent pitcher. Kanzi was prompted to watch as each was poured. Then, the experimenter pretended to pour one cup back into the pitcher, and subsequently removed the pitcher from the table. When asked which cup contained the liquid, Kanzi would point, and his response was recorded. His answers indicated whether he was able to accurately determine where the imaginary tea ended up.

Kanzi selected the correct cup in the majority of trials, demonstrating high-level cognition not typically expected in apes. However, scientists were still unable to confirm whether he truly understood pretend play, or if he thought there was real liquid poured into the cup. The researchers eliminated this possibility in a second control experiment by comparing a cup of actual juice to a cup of pretend juice. The experimenters placed a cup full of real juice on the table, and then pretended to pour juice into a second cup. When asked which one he wanted, Kanzi most often chose the cup with actual juice, verifying that he could distinguish between real and pretend juice, and thus indicating that he truly was able to conceptualize the make-believe juice in the first experiment.

The experimental evidence from these tests supports the possibility that nonhuman animals can track pretend objects across multiple steps by a partner in pretend play. However, Kanzi isn't just any ape. The researchers were able to evaluate Kanzi due to his existing ability to respond to verbal prompts and engage in the game through pointing. Therefore, the effect of Kanzi's language training on his performance is a critical consideration, and it is uncertain whether these results can be generalized to other apes.

Studies that explore primate cognition continue to challenge the foundations of how we classify humans versus animals. The development of theory of mind, or the ability to understand others' perspectives, is a critical step in childhood development. Theory of mind has long been considered to be a uniquely human attribute, separating our species from the rest of the animal kingdom. However, animal studies are building increasing evidence that we share many mental processes with our evolutionary ancestors.

"Previously featured in *The New York Times* and on *The Oprah Winfrey Show*, celebrity bonobo Kanzi has taken the spotlight once again."

Future research will need to explore whether other apes and mammals can produce similar behaviors, and furthermore, replicate results with greater sample sizes to rule out luck, chance, and an educational advantage in Kanzi's results. With further verification, scientists could one day confirm that nonhuman animals are capable of imagination and thinking outside of reality, evolving our understanding of primate cognition and challenging the roots of what it means to be human.

Cell Metabolism (2024). DOI: 10.1016/j.cmet.2024.05.012
Institute for Quality and Efficiency in Health Care (2022). PMID: NBK279454

PHOTO VIA ADOBE STOCK

OVERHEARD AND UNDERSTOOD

Word learning in gifted dogs

BY AMELIA DSOUZA, DATA SCIENCE & BEHAVIORAL NEUROSCIENCE, 2026 & KATRINA LEE, HEALTH SCIENCE, 2026
DESIGN BY CLAIRE HANSON, HEALTH SCIENCE AND ENTREPRENEURSHIP, 2027

Dogs have long been known as man's best friend, valued for their loyalty, companionship, and unconditional love. Beyond this bond, they are increasingly recognized as extremely intelligent animals capable of interpreting human words and gestures. Human language is uniquely complex, serving as a primary tool for interpersonal communication and social interaction, and is acquired through constant exposure in social settings. By just 18 months of age, toddlers develop new vocabulary by passively observing other people's interactions. Research has found that dogs are also able to learn words through overhearing, similar to toddlers. A select group of domestic dogs, known as Gifted Word Learner (GWL) dogs, possess a "sociocognitive toolkit" similar to that of an 18-month-old infant. These canines don't just follow commands; they are capable of learning names and phrases by simply eavesdropping on human conversations.

To explore the depths of their talents, researchers tested whether dogs could learn through passive observation, an accomplishment that in humans requires sophisticated social skills and understanding of communicative intent. In the primary experiment, ten GWL dogs were tested under two conditions: "Addressed," where the owner spoke directly to their dog, and "Overhearing," where the dog observed two owners discussing and passing a toy back and forth without acknowledging the dog. The exposure protocol involved a structured schedule over a couple of days that consisted of one minute of labeling the toy, followed by three minutes of play without mentioning the label, and up to twenty minutes of free exploration of the toy. The dogs performed significantly well in both conditions, with no statistical difference in their proficiency between the two conditions. This suggests that GWL dogs do not need direct social interaction with their owner to form object-label mappings.

Researchers noted that temporal contiguity between the labeling action and presentation of the objects may have affected the observed success in the GWL dogs' label learning. In other words, timing, specifically how closely in time the spoken label of the toy occurs compared to the object, may make it easier for the dogs to connect the word to the object. Another experiment was conducted to analyze how timing affects label learning in GWL dogs. The procedure replicated the schedule of the exposure protocol in the primary experiment, but with a key modification. During the one minute of labeling, the owner showed the dog a new toy, placed it into a bucket, then let the dog take it out before putting it back inside. The bucket was then lifted so the dog could no longer see the toy. The owner said the toy's label in short sentences while the toy was hidden and looked back between the toy and the dog. Overall, the dogs performed well when asked to identify the new dog toy labels. The dogs performed equally well even after conducting the same experiment two weeks later, suggesting that the object-label associations were retained over time.

Interestingly, this phenomenon isn't shared by the average family dog. Border Collies, a breed known for their intelligence, were tested using the same protocols and failed to learn new labels. While they more often than not picked the correct new toy, researchers concluded that this was due to "neophilia", or a strong affinity for newness and novelty rather than word comprehension.

The overall findings suggest that the ability to learn through overheard and eavesdropped speech is an idiosyncratic trait limited to GWL dogs. Notably, this hints that cognitive skills accompanying language acquisition are not exclusive to humans. Rather, they could rely on broader evolutionary mechanisms that precede language itself. In the shared ecological niche of humans and canines, some "good boys" have evolved to be very good listeners.

Science (2026). DOI: 10.1126/science.adq5474

PHOTO VIA ADOBE STOCK

THE HALF-LIFE OF A HEALTHY BRAIN

Proteostasis and aging

BY IBA BAIG, DATA SCIENCE & BEHAVIORAL NEUROSCIENCE, 2026
DESIGN BY GIULIA WALKER, DESIGN, 2026

Unlike most of the body's cells which regularly divide, die off, and renew themselves, neurons stay from birth until death. The process of keeping them healthy for a lifetime is extremely important, and the system responsible for this is called proteostasis, or protein homeostasis. Scientists attribute the natural aging process and disease pathology to gradual faults in this system. Understanding why it breaks down is one of the most pressing questions in neuroscience.

Proteins are the molecular building blocks that are only useful when correctly folded into precise three-dimensional shapes. Misfolded proteins are not just useless, but they can be actively toxic and aggregate to disrupt cellular machinery. To prevent this, cells maintain three defenses: molecular chaperones that assist protein folding, the ubiquitin proteasome system (UPS) that destroys damaged proteins tagged for disposal, and autophagy which engulfs large debris for breakdown. Together these form the proteostasis network (PN). Activity shifts across tissues and life stages, and this complexity has profound consequences for disease.

Not all cells face equal proteostatic pressure. Dividing cells dilute their burden of damaged proteins with each split, passing a fraction of cluttered damage to daughter cells. Neurons cannot do this since they are post-mitotic and must clear accumulated proteins through the PN with no reset. Some neurons extend axons over a meter in length, meaning proteins synthesized in the cell body must travel vast distances to synaptic terminals and back. These synaptic proteins are especially at risk, recycled far from the cell's protein-making machinery and among the first casualties when the system falters.

All three stages of the PN decline with age: chaperone availability drops, proteasome activity slows, and autophagy becomes less efficient. The consequences appear in some devastating conditions such as amyloid-β and tau aggregates in Alzheimer's disease, □-synuclein Lewy bodies in Parkinson's, and TDP-43 deposits in ALS. In a 2016 study, a team of Stanford researchers used engineered mouse models that allow cell-type specific tracking of neuronal proteins. The team found that neuronal protein half-life approximately doubled between young adult and aged mice. The same study catalogued an "aged neuronal aggregome" of 1,726 proteins found in aggregates in aging brains, half of which showed reduced degradation with age. Many were familiar disease-associated proteins; hundreds had never been linked to neurodegeneration before. Synaptic proteins were overrepresented throughout.

Proteostasis has long been viewed as each cell's private responsibility, but emerging research is rewriting that assumption. Microglia — the brain's resident immune cells — actively engulf neuronal proteins, acting as a cellular waste-removal service. The Stanford study found that over half of the neuronal proteins accumulating in aged microglia already showed signs of proteostatic failure: slower degradation or aggregation. Meanwhile, a UCBerkeley study in C. elegans found that triggering a mitochondrial stress response in an astrocyte-like glia was sufficient to reduce protein aggregation in neurons, with protective signals relayed via small vesicles. Proteostasis in the nervous system, it turns out, is a collaborative enterprise.

Caloric restriction and exercise both upregulate autophagy with extended lifespan in model organisms by boosting the same pathway. HSP70-activating chaperone inducers and modulators of the integrated stress response are under active investigation. None have yet been proven to restore neuronal proteostasis in aging humans, but the field is advancing rapidly. A fuller understanding of how and when the network fails will be essential for developing therapies that keep the aging brain's protein quality control running for as long as possible, and maybe even one day reverse aging itself.

Keeping the neuronal desk clean is the molecular foundation of a healthy brain across a lifetime. When the system fails, the consequences range from subtle cognitive slowing to outright neurodegeneration or disease. What is now clear is that this failure is not a neuron-level problem alone, it involves the coordinated breakdown of multiple cell types over decades. Restoring proteostatic balance may be one of the most tractable paths to preserving brain health as we age.

Nature (2026). DOI: 10.1038/s41586-025-09987-9
Science Advances (2023). DOI: 10.1126/sciadv.adi1411
Annu Rev Biochem (2015). DOI: 10.1146/annurev-biochem-060614-033955

PHOTO VIA ADOBE STOCK

Silent reshaping

How menopause affects the brain

BY SOPHIA CHENG, PUBLIC HEALTH, 2029

DESIGN BY CANDICE MEDINA, MECHANICAL ENGINEERING, 2026

Most women will experience menopause in their lifetime, but its effect on the brain remains surprisingly understudied. New research from the University of Cambridge is changing that. It revealed that menopause has a hand in quietly reshaping brain structure, mood, and cognition in ways that carry long-term consequences for women's health.

> **"The brain regions where we saw these differences are ones that tend to be affected by Alzheimer's disease. Menopause could make these women vulnerable further down the line."**
>
> — Professor Barbara Sahakian, University of Cambridge

In a study analyzing nearly 125,000 women from the UK Biobank, researchers at the University of Cambridge found that post-menopausal women showed drastic reductions in grey matter volume in many critical brain regions. Grey matter contains nerve cells that are essential for processing information, supporting memories, and regulating emotions. The most affected areas were the hippocampus, which is responsible for storing memories, the entorhinal cortex, which is a gateway between the hippocampus and the rest of the brain, and the anterior cingulate cortex, which helps manage emotions and attention. In addition, a literature review in 2025 from the Annual Meeting of The Menopause Society confirmed that multiple studies have documented gray matter reductions in the frontal and temporal cortices as well, which are critical for memory and executive function.

These affected brain regions are also common to the progression of Alzheimer's disease. This overlap is noteworthy given that nearly two-thirds of Americans living with Alzheimer's are women. While women having a longer life expectancy has been cited as the primary explanation for this disparity, many researchers are looking beyond this seemingly simple explanation. Cambridge psychiatry professor Barbara Sahkian explained that menopause could make women more vulnerable to neurodegeneration later in life, helping to explain the disproportionate burden of dementia among older women. Stephanie Faubion, medical director for The Menopause Society, stated that this data will hopefully lead to a better understanding of the cognitive concerns women experience during menopause, so that researchers can ultimately identify effective therapies.

Christelle Langley, a researcher in the Department of Psychiatry at the University of Cambridge, stated that a healthy lifestyle, including exercise, staying active, and eating well are important things to maintain during menopause to help mitigate some of its effects. However, she also stressed that a broader cultural shift is needed for there to be more understanding about the physical health as well as the mental health of women going through menopause. As researchers continue to uncover the connections between hormonal changes and brain structures, the need for targeted therapies and earlier intervention becomes more immediate. Closing the gap in menopause research is more urgent than others believe. Due to the disproportionate burden of Alzheimer's disease among women, it is imperative scientists dedicate more research towards menopause in order to ensure that women's cognitive health is taken seriously in every stage of their lives.

Psychological Medicine (2026). DOI: 10.1017/S0033291725102845

MULTIPLIERS OF HUMAN CAPABILITY

INVENTIONS THAT CHARTED THE WAY TO AGI

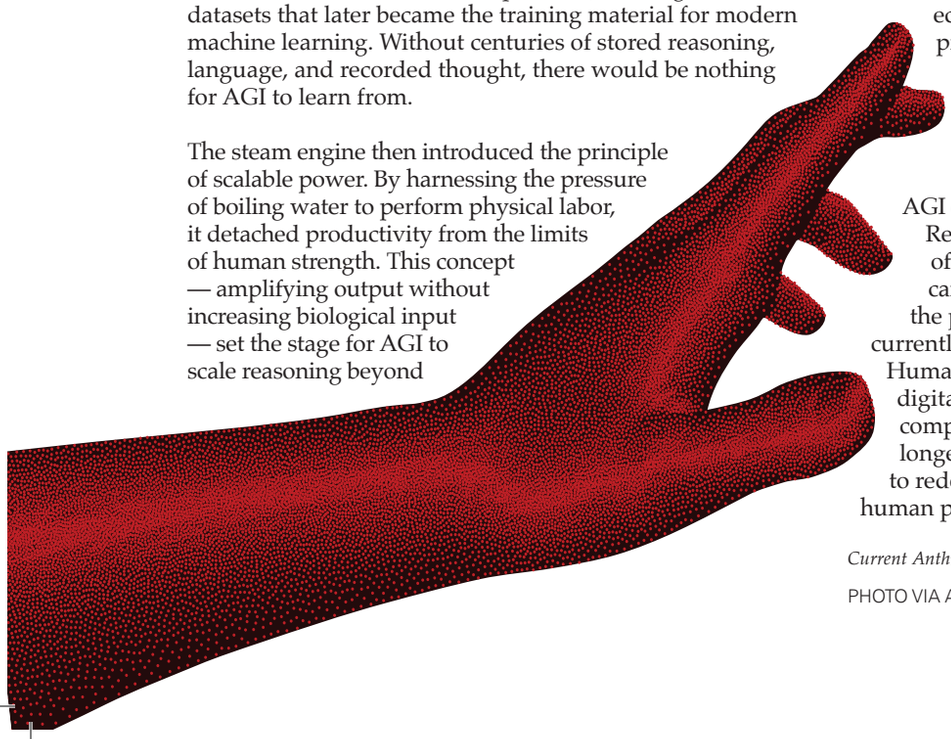
BY IBA BAIG, DATA SCIENCE & BEHAVIORAL NEUROSCIENCE, 2026
DESIGN BY GIULIA WALKER, DESIGN, 2026

Since the beginning of life on Earth, humanity has possessed a unique skill that sets it apart from other species: the ability to create and use tools to interact with the world. Innovation isn't just a list of gadgets; it is a consistent process of offloading biological limitations — digestion, memory, muscle, and now cognition — onto external systems. The long-held dream of a 'universal tool' capable of solving any problem is no longer as far-fetched as it seemed a century ago. This is the promise of Artificial General Intelligence (AGI): a technological mind that doesn't just follow instructions, but learns and thinks with the broad flexibility of humans, capable of understanding, adapting, and generating solutions autonomously.

The first major step toward AGI was fire. By allowing humans to cook food, fire outsourced the energy cost of digestion, thereby diverting excess calories to brain development. Fire was the biological prerequisite for AGI; it shaped a brain capable of designing technological intelligence. Fire marked the first instance of humanity systematically transforming biological constraints into external systems — a pattern that would repeat with each subsequent breakthrough.

The next leap came with writing, which externalized memory in a permanent medium. Before writing, knowledge died with the individual or was distorted through oral communication. Writing enabled the exchange of information globally. It serves as an accumulation of the human experience, creating vast datasets that later became the training material for modern machine learning. Without centuries of stored reasoning, language, and recorded thought, there would be nothing for AGI to learn from.

The steam engine then introduced the principle of scalable power. By harnessing the pressure of boiling water to perform physical labor, it detached productivity from the limits of human strength. This concept — amplifying output without increasing biological input — set the stage for AGI to scale reasoning beyond the limits of a single mind. Just as the industrial revolution multiplied physical labor, AGI aims to multiply cognition.



Arguably, the transistor was the most important piece of the puzzle that would lead to AGI. By replacing bulky vacuum tubes with microscopic electrical gates, transistors allowed computers to flip a switch between 'on' and 'off' states to perform billions of operations per second in tiny, energy-efficient packages. These gates proved that complex reasoning could be encoded in physical systems. Layered together, they created machines capable of learning and navigating the world. Finally, the necessary material conditions were in place for AGI to exist.

Each of these breakthroughs was more than an isolated invention; they were milestones in a consistent journey: finding a human limitation and creating a system to solve it. Fire expanded the brain, writing preserved knowledge, the steam engine scaled power, and transistors enabled the speed of thought. Collectively, they illustrate a persistent human strategy: offload constraints to extend capability.

Unlike Narrow Artificial Intelligence (systems like GPS or chatbots that are brilliant at performing a single task but blind to everything else), AGI represents a fundamental shift. It does not merely follow instructions, but understands, learns, and evolves independently. It replicates general reasoning rather than extending a single human function, making it historically unprecedented.

AGI can integrate complex variables — data, economics, ecology, politics — simultaneously, producing solutions beyond the reach of any individual expert. In this sense, it is the culmination of millions of years of offloading, from muscle, to memory, to machinery, and now to reasoning itself.

AGI challenges traditional notions of intelligence. Reasoning is no longer confined to the boundaries of a brain; logic, creativity, and problem solving can exist as mathematical functions. It holds the potential to help navigate problems that are currently too complex for humans to solve alone. Humanity is on the cusp of collaborating with a digital intellect capable of tackling problems too complex to solve alone. The universal tool is no longer hypothetical; it is emerging in reality, ready to redefine the limits of intelligence, autonomy, and human potential.

Current Anthropology (2017) DOI: 10.1086/692113

PHOTO VIA ADOBE STOCK

Digging up old wounds

New findings on the Hjortspring boat

BY CECELIA CLEMENTE, CHEMICAL ENGINEERING, 2028
DESIGN BY CECELIA CLEMENTE, CHEMICAL ENGINEERING, 2028

Despite centuries of seafaring and excavation in Scandinavia, the Hjortspring boat is the single, complete prehistoric sewn plank ship that has been discovered in the region. About 40% of it was extracted from the bog when it was discovered in the 1920s. Scientists quickly moved to preserve it using alum and polyethylene glycol, which limited the analysis that could be done on the pieces to both date and locate the boat's origins. In 2024, scientists decided to reexamine the boat, which led to the confirmation of existing theories and new hypotheses, prompting future studies.

When it was first extracted, the boat was believed to be from the third or fourth century BCE due to rock art of boats that were very similar in style to the Hjortspring. Scientists were able to return to the bog where the Hjortspring was first discovered, and they found wood, which was later subject to carbon-dating in order to figure out the time period it originated from. This data supported the previously believed date. Location-wise, the weapons found aboard matched much of northeast Europe from that time.

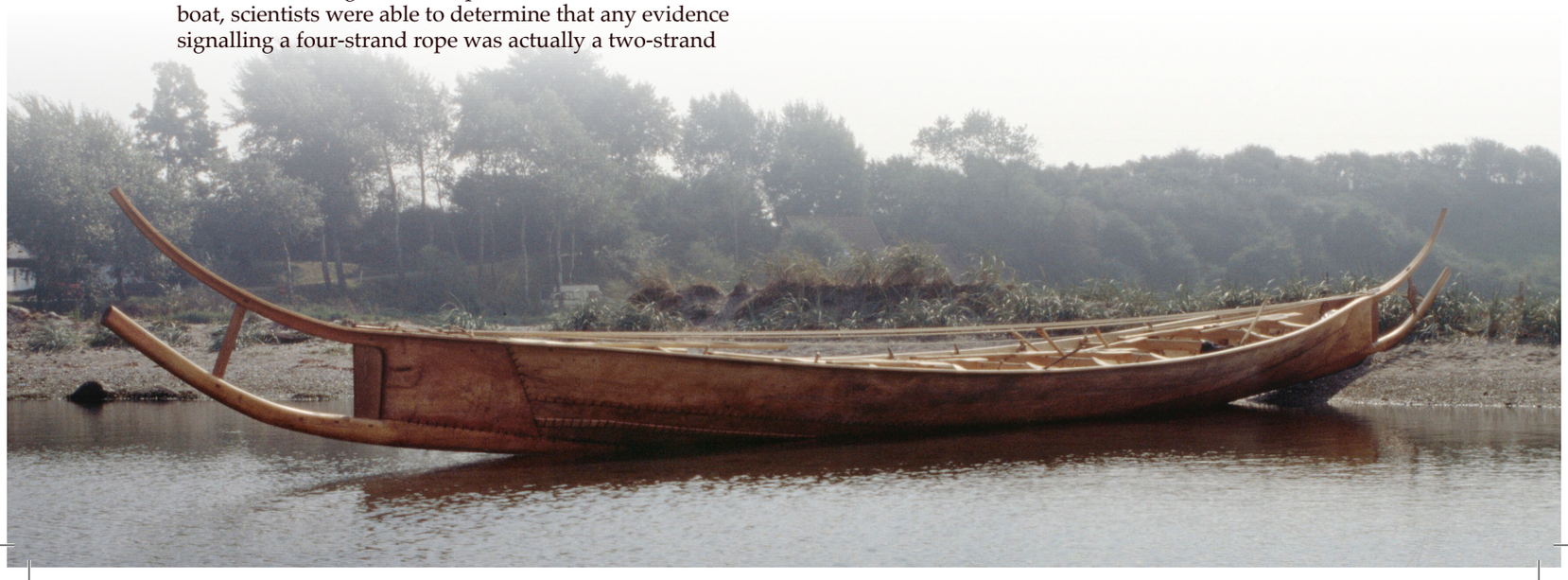
For decades, these vague theories were all that was known about Hjortspring. However, in 2024, new scientists approached the problem equipped with modern scientific techniques and fresh perspectives to investigate Hjortspring. They looked at caulking and cordage, which had been excavated with the ship, but went unexamined and unpreserved. The cordage used to tie the boat's planks together was 3D modeled and analyzed using carbon-14 dating. This dating verified the third or fourth century BCE dates previously theorized. Additionally, the cordage was examined for how it was built. The pieces are two strands of thinner rope twisted together. A previously unanswered question about Hjortspring was how and why it had evidence of ropes with a variety of strand numbers. After reexamining the techniques used to construct the boat, scientists were able to determine that any evidence signalling a four-strand rope was actually a two-strand rope. The two strands had been either spliced together to lengthen it or tied into knots, which gave the look of four strands in the imprints it left in caulking material.

They also studied the caulking used to waterproof the boat. Using a special analytical technique called gas chromatography-mass spectrometry, it was determined that the caulking could be made of glycerol, palmitic acid, stearic acid, and maybe lard. To locate the boat's origin, they compared the mass spectrums of the caulking and birch tar. Due to the lack of similarities between the two spectrums, it was determined that the caulking was made of coniferous tar, likely pine, and fat. The presence of pine narrows the region of the boat down to Blekinge, Gotland, north Poland, or Bornholm. The rock art from these regions matches the style of Hjortspring and continues to support the likelihood of these regions being the boat's origin. Additionally, a partial fingerprint was discovered in the caulking. Much cannot be gleaned about the person who left the fingerprint, but it is likely they were one of the sailors aboard the ship, not just a shipbuilder.

Even though there were no newly discovered pieces or revolutionary changes, reexamining the Hjortspring boat with current technology and fresh eyes allowed many more discoveries regarding its dating and location. A partial fingerprint was found, and while there isn't any current information about its creator, it offers an incentive for reexamination. These new findings are crucial to fully placing the Hjortspring in history, but should not be viewed as fully conclusive. Just as Hjortspring's first investigation was for this study, the findings should be viewed as a starting point and encourage continuous evaluation of old beliefs and new research methods.

Public Library Of Science (2025). DOI: 10.1371

PHOTOS VIA ADOBE STOCK & WIKIMEDIA COMMONS



JEALOUSY IN THE JUNGLE

BY RISHMITA RAO, PHARMACY, 2031
DESIGN BY DIVYA GANDHI, CHEMISTRY, 2029

Imagine fighting with a sibling over who's going to get the last slice of cake from the fridge. Now, imagine two monkeys fighting over affection from their mother. Similar, right? This shared feeling is *jealousy*. Monkeys tend to form relationships and attachments with their counterparts and feel distressed when separated from each other. So, do monkeys feel jealousy like humans do?

Many different studies have been conducted to see if primates feel jealousy. In 2017, scientist Karen Bales conducted an experiment in which she and colleagues separated male Titi monkeys from their female partners, putting a female Titi in a room with a random male Titi for 30 minutes. The separated male Titis' behavioral and hormonal responses were recorded. The monkeys' testosterone and cortisol levels increased, which indicated a rise in stress. These changes correlated directly with the amount of time that the male monkeys watched their partners with the stranger male. The rise in testosterone was expected given its association with mating-related aggression. In another study from 2021, a team of scientists in Namibia followed chacma baboon families in a nature park. Adolescent baboons would attempt to divert their mother's attention from their siblings, sometimes using tricks to lure their siblings away from their parents. Furthermore, much of this behavior occurred during grooming, which involves the removal of bugs and dirt from a monkey's fur. Grooming is known to strengthen social bonds, as hugs do for humans. So, these observations demonstrate jealousy's role in bonding and socialization.

These related studies convey how monkeys feel jealousy in similar ways to humans, which can be attributed to evolutionary factors. Jealousy motivates humans differently than other emotions, and if humans feel it, then it may have helped related ancestors form bonds and relationships in a similar way. Jealousy exerts a powerful influence on behavior, different from other emotions like anger or sadness. It motivates individuals to pay closer attention to their feelings and repair bonds.

> **"Jealousy exerts a powerful influence on behavior, different from other emotions like anger or sadness."**

Jealousy in monkeys often reflects sibling rivalry in human relationships. Children tend to get upset when a parent shows favoritism for another sibling over them, and might seek their attention by making a mess or screaming. Monkeys show similar behaviors when competing for parental affection. Additionally, jealousy might serve as a survival tactic for both species. When humans feel jealous, it is because they are trying to protect themselves from internal insecurity and fear. This is reflected in monkeys, because they may feel a sense of abandonment when their sibling is being nurtured over them. Maintaining relationships is important for food access, learning how to live on their own, and finding their own mate.

As demonstrated by analyzing primates, our closest evolutionary ancestors, jealousy is an age-old trait that lays the foundations for motivation, relationships, and survival.

Proceedings of the Royal Society (2026). DOI: 10.1098/rspb.2025.2102
Frontiers in Ecology & Evolution (2017). DOI: 10.3389/fevo.2017.00119

Release the Kraken!

The mystery of the giant squid

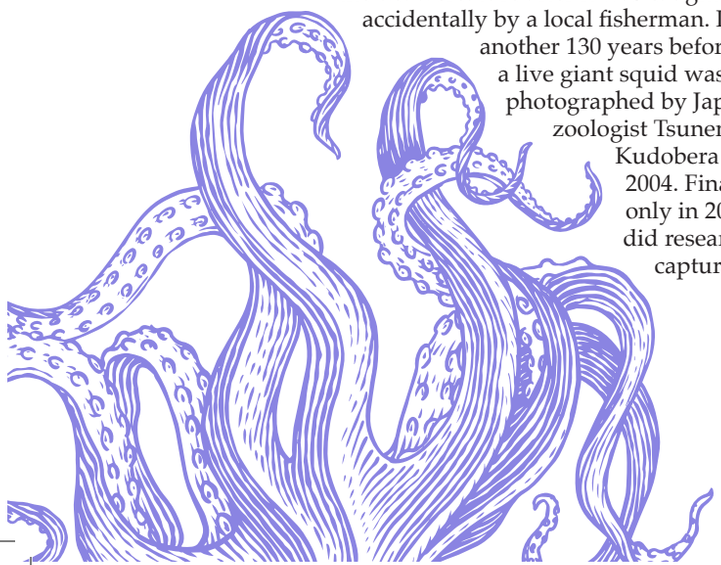
BY CAROLINE GABLE, HEALTH SCIENCE & PSYCHOLOGY, 2026
DESIGN BY CANDICE MEDINA, MECHANICAL ENGINEERING, 2026

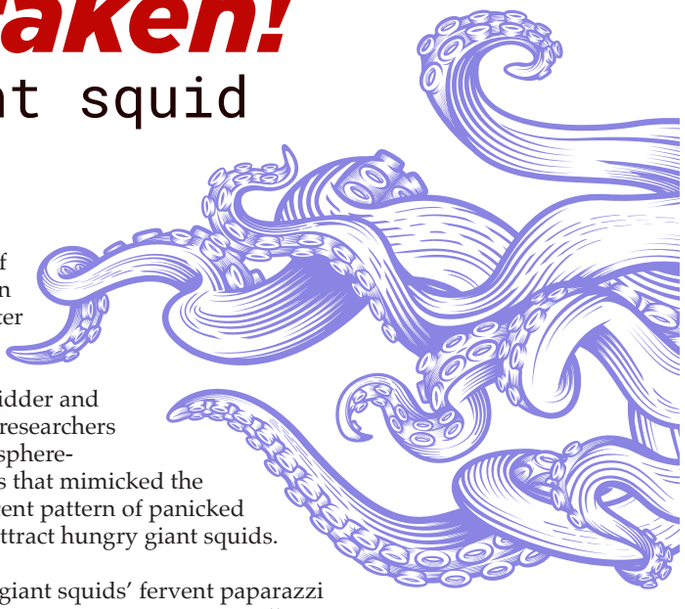
Whispers and tales of the "*kraken*" could be heard around the campfires of Scandinavian sailors and Vikings as early as the 12th century. Derived from Old Norse, the root word "*krake*" roughly translates to a "malformed or overgrown, crooked tree" and was originally used in reference to wooden Scandinavian boat hooks, drags, or anchors. The image of these twisted anchors was the closest resemblance sailors of the time had to the sea monster of enormous, ship-devouring, cephalopod-like proportions they claim haunted the ocean on their voyages. A monster they claimed conjured whirlpools and dragged boats down into the abyss.

These legends were largely discounted as Scandinavian mythology and folklore in the mid-1800s, when discoveries of kraken-like creatures now known as *Architeuthis*, or giant squids, began on land and in water. However, giant squids appear to be just as shrouded in mystery as the legendary kraken.

The giant squid is a massive, deep-sea cephalopod that is estimated to grow up to 60 feet long and weigh close to 500 pounds. With eight arms and two long tentacles used for hunting, giant squids also have three hearts, a beak, and the largest eyes in the animal kingdom. Their eyes measure 10-12 inches in diameter, roughly the size of a beach ball, and allow them to see bioluminescent light. These cephalopods live worldwide in the mesopelagic and bathypelagic zones of the ocean, as deep as 3,300 feet. It is primarily due to this depth that so much remains unknown about giant squids, and most research relies on their corpses being found in the stomach of sperm whales or washed up on beaches.

In fact, the first photo of a giant squid was taken in Newfoundland in 1874, when a Reverend paid $10 for the dead creature after it was caught accidentally by a local fisherman. It took another 130 years before a live giant squid was photographed by Japanese zoologist Tsunemi Kudobera in 2004. Finally, only in 2012 did researchers capture the first video of the animal in its deep-water habitat, after Dr. Edith Widder and her team of researchers engineered sphere-shaped lures that mimicked the bioluminescent pattern of panicked jellyfish to attract hungry giant squids.





Despite the giant squids' fervent paparazzi and population estimates over 100 million, sightings of these beautiful beasts are rare. And perhaps it should stay that way.

The allure of debunking ancient legends is strong, but it is critical that human contact with these giant squids is limited… and quiet. In 2011, Professor Michel André and his research team found that sound from industrial human activities, such as ocean exploration, causes lesions in the statocysts of cephalopods. Statocysts are fluid-filled organs that allow giant squids to maintain balance and orientation and are crucial to their survival. Consistent with these findings, the statocysts of nine giant squids discovered dead off the coast of Spain between 2001 and 2003 were severely damaged. Their deaths were later linked to seismic surveys and Spanish Navy sonar tests.

Luckily, giant squids are safe from becoming human prey, unlike their smaller cephalopod counterparts in shallower water. Giant squids contain a high concentration of ammonium chloride that helps them be neutrally buoyant in sodium chloride-rich deep seawater. This ammonium has led unfortunate taste-testers to describe raw giant squid as "rotten, salty licorice" that smells like cat pee. Therefore, giant squids are both difficult to study and unappealing appetizers, factors that may protect them from excessive human contact and allow many of their secrets to remain hidden.

As researchers learn more about giant squids, it is fair to say that myth has slowly become reality. The terrifying creature of folklore may actually not be as ship-wrecking and man-eating as previously thought by Norse sailors, but rather, a fragile inhabitant of a largely untouched home.

História, Ciências, Saúde-Manguinhos (2014). PMID: 25338036
The Royal Society (2005). DOI: 10.1098/rspb.2005.3158

PHOTOS VIA ADOBE STOCK

Seawater draws streaks and loops against pebbles at Castle Island, Boston.

PHOTO BY JIAJIA FU, BIOENGINEERING, 2026

The sunset reflects off the car at Wilcox Playa, Arizona. Cranes flock to their evening resting site just out of sight.

PHOTO BY JIAJIA FU, BIOENGINEERING, 2026

One step forward, one step back

The stagnant gender pay gap

BY ANGALINA COX, CELL & MOLECULAR BIOLOGY, 2028

DESIGN BY CECELIA CLEMENTE, CHEMICAL ENGINEERING, 2028

In 1995, an equal percentage of men and women completed their bachelor's degrees. Since then, the gap has expanded with women in the lead. In 2024, 47% of young women and 37% of young men aged 25 to 34 held a bachelor's degree. This trend is consistent across all major ethnic and racial groups, and yet correlating wages fail to follow suit. While women increasingly dominate higher education, recent polls find they continue to be paid only 83 cents on a man's dollar.

According to an analysis created by the Pew Research Center, 50.7% of the labor force that is college-educated in the United States is made up of women. Women first eclipsed men as the majority in 2019, and the trend has continued since. The increase in earnings gained from a college degree is known as the college wage premium. The declining growth of this premium lends a possible explanation for the slowing pace of achieving gender pay parity. This does not solve the puzzle, however, as the gender pay gap remains very similar for all women, no matter their level of education.

Since 2002, the gender pay gap has stalled at a 17% difference between what women and men earn as full-time workers within the United States. Young workers aged 25 to 34 see only a 5% difference, but the gap grows as adults progress. In 2010, for example, young women gained 92% of what their young male counterparts earned, but a decade later, this demographic of women was found to earn 84% in comparison to men of the same age. This pattern of young women facing a smaller pay gap, only to see that gap widen as they age, was found to occur across multiple generations in the 2000's.

A dominant influence contributing to this disparity is likely entering parenthood. The greatest increase in the pay gap happens within the 35 to 44 age group, which reflects the period in which women are most likely to raise young children at home. In direct contradiction, employed men in the same age group assumed to have young children have been found to receive greater pay. These pay fluctuations are known as "motherhood wage penalties" and "fatherhood wage premiums."

The expansion of the pay gap for parents in the workforce is therefore not caused by mothers receiving less for being less qualified, but rather is caused by fathers experiencing an increase in their pay. Mothers are not being financially punished for having children, but they are not being rewarded the way fathers are.

Paired with aggressive return to office policies that have been leaving women who act as caretakers with the choice of finding childcare or leaving employment altogether, the pay gap has further stagnated.

Statistically, more women are going to college, completing their degrees, and entering the workforce on a strong footing, but reaching gender pay parity is still out of grasp. The gap is not predicted to close until 2056 for all women on average, with women of color further disadvantaged.

Demography (2018). DOI: 10.1007/s13524-018-0712-5

PHOTOS VIA ADOBE STOCK

More than just a beat

The science behind house music

BY DANIELLE JEONG, CHEMICAL ENGINEERING, 2028

DESIGN BY CECELIA CLEMENTE, CHEMICAL ENGINEERING, 2028

Chances are, names like Calvin Harris, John Summit, and Dom Dolla are already familiar. Their sets at festivals like Lollapalooza, Ultra, and EDC have become can't-miss events, and what ties them all together is house music. Unlike a typical song with verses and choruses, house music is built on steady beats, deep basslines, and repeating patterns that layer on top of each other. The genre has been filling dance floors for over four decades, but what is it about repetitive music that people can't get enough of?

House music doesn't tell a story the way most songs do. Instead, it builds an experience. A track might start with just a drum beat, then gradually add a bassline, then synths, then vocals — stacking sounds on top of each other until the energy peaks at what's called the "drop," the moment that keeps people chasing the next song. A 2024 study by Wulf and Kitahara analyzed 194 house tracks and measured volume, frequency, and drum patterns across different sections of each song. They found that the drop is not just perceived as more intense, it measurably is.

The drums and bass pump out more sound energy during the drop compared to any other part of the track. That's why the drop doesn't just sound loud, it *feels* physical, as if the bass is hitting right in the chest. Interestingly, the same study found that while the volume changes dramatically between sections, the overall tone of the music stays remarkably consistent throughout. That balance may explain why house music can feel both energizing and almost meditative.

Musicologist Stan Hawkins explored this push and pull between sameness and change by closely analyzing Lil' Louis's 1989 hit "French Kiss." He found that house tracks are built around repeating groove cycles that slowly transform as the song progresses. The listener stays hooked on the rhythm because the core beat never changes, but their attention is held because something is always subtly shifting.

Similarly, Wulf and Kitahara found the thumping pulse at the center of every track was maintained at an almost identical rhythm across all 194 songs. The kick drum serves as the anchor, and everything else moves around it. That central pulse typically lands around 120 to 130 beats per minute, which happens to match an elevated heart rate during physical activity. This means the song is essentially syncing the music to the body's own rhythm.

The sound alone doesn't fully explain house music's hold on people. The genre was never meant to be listened to alone. It was made for a room full of people moving together.

Hawkins explains that the dancefloor creates a temporary community where everyone syncs to the same beat. This sense of togetherness goes back to the genre's origins. As historian Brian Lindgren states, house music was born in the early 1980s in clubs intentionally built as welcoming spaces for people of different backgrounds. Independent producers made music in their bedrooms, giving the music a sense of authenticity.

House music isn't just background noise at a party; it is a carefully constructed experience. The sound triggers a physical response, the structure keeps the brain hooked through subtle changes, and the crowd transforms it into a collective moment that's hard to replicate anywhere else. And the genre isn't standing still.

House surged to the second most downloaded music genre in 2025. Much of that growth has been driven by subgenres like Afro house and artists like Peggy Gou and Fred Again, who are blending classic house grooves with Afrobeat rhythms, UK garage, and trance-inspired melodies. Dance and electronic music now account for 45% of Coachella's 2026 lineup, and artists are now headlining their own festivals. So the next time a John Summit track comes on and the whole room starts moving together, it's not just a good song — it's science, culture, and community all wrapped up in a beat.

2024 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (2025) DOI: 10.1109/APSIPAASC63619.2025.10849244
Analyzing Popular Music (2003). DOI: 10.1017/CBO9780511482014.005

PHOTO VIA ADOBE STOCK

The illusion of RECYCLING

BY MARIA BATTLE, PSYCHOLOGY & NEUROSCIENCE, 2026

DESIGN BY CLAIRE HANSON, HEALTH SCIENCE & ENTREPRENEURSHIP

We've all heard it since kindergarten: "Reduce, reuse, recycle." A great mantra, but not quite enough anymore, and really, it never was. Recycling, while well-intentioned, is often co-opted by corporations as a form of greenwashing that shifts responsibility onto consumers while avoiding real systemic change.

Currently, the majority of recycling is thrown in with the trash. In the US, only 32% of materials that are "recycled" actually end up going through the process of being recycled. In Boston, this number falls to 19%, and globally, a staggering 9-10%.

Locally, at Boston College, trash and recycling bins are set out at each catered event, encouraging people to use them properly. However, working at the school, I observed that the trash and recycling just get mixed together. This reveals how the appearance of sustainability is often more important than the actual impact, and how the public is made to feel guilty while larger powers are not pulling their share. Boston College is not the only institution to use recycling as a form of greenwashing — a way to appear morally responsible without making real changes. But seeing it firsthand was a disappointing moment, and it showed that a college that proclaimed to emphasize service and ethical development just wanted to put on a show for the people who filled their pockets. This experience made it clear that the current system of recycling is deeply flawed, and often just a feel-good illusion. Instead of relying on broken systems, we need to push for practices that actually make a difference, like reducing consumption and holding institutions accountable for their environmental claims.

Let's take it back to a different time in history, before we had the knowledge we possess today. Ironically, plastic was initially invented to try and reduce the level of negative impact humans had on the environment. Early versions of plastic were used to replace the ivory used in billiard balls, and, in theory, curb the risk posed to elephants due to the growing ivory demand in the 1860s. Fast-forward to the World Wars, and there was a massive demand for plastics to replace scarce materials like silk, rubber, and metal. This is when the world was introduced to new plastics like nylon and polyethylene. Polyethylene would go on to become the key material for items like shopping bags, plastic films, milk jugs, and shampoo bottles. Nylon was incorporated into things like ropes, food packaging, tires, toothbrushes, combs, and so much more. Soon, these items became a part of daily life.

The convenience and ease of plastic was undeniable, and still is. Advertisements for plastic highlighted its durability and affordability. The popularity of plastics reached an all-time high in the 1950s and '60s, particularly in 1956, when influential figures encouraged the plastics industry to shift campaign ads toward promoting "throwaway living" in an effort to boost sales. Coincidentally, the 1950s were also when scientists became aware of the detrimental effects of plastic pollution. This is when they first linked human-made chemicals to air pollution. They also started to notice plastic pollution in oceans around the same time that oil-based plastics became common in consumer goods. Unfortunately, the plastic industry is well aware of the harm they impose, and even more tragically, they have been aware of these implications since at least the beginning of the 1970s, and realistically, since the 1950s.

The public is becoming more and more aware of the facade that corporations hide behind; even so, the plastics industry continues to deny the severity and widespread impact of mass plastic pollution. This industry, in pursuit of financial gain, has opposed sustainable solutions, even going so far as to fight local regulations of disposable plastic products. They have, at times, acknowledged the problem, but they continue to shift blame onto public consumption and claim they can only be held accountable for small parts of waste while the rest is "beyond their control."

By the 1960s and '70s, it was becoming abundantly clear that plastic was not as great a solution as it seemed. There was growing public concern about the overflow of waste in landfills, air pollution, visible litter in cities, water sources, and on highways. The scientific community continued to publish more findings about the negative effects of plastics and increase public awareness. On April 22nd, 1970, the US held its first-ever Earth Day. This is considered to be the spark that ignited environmental movements throughout the '70s. Then, in 1997, a mariner sailing from California to Hawaii discovered the Great Pacific Garbage Patch. At this point, the public could no longer turn a blind eye to the problem. This allowed for the issue of pollution to be viewed less abstractly and be seen for what it really is: a large-scale, tangible crisis that affects every living organism on the planet. It's important to acknowledge that although it took the world a long time to reach this point, this global shift in paradigm was a critical juncture.

"Good intentions are manipulated into a smokescreen — one that obscures the urgent need to hold corporations truly responsible."

As environmental awareness grew, millions of well-meaning people embraced the mantra to "reduce, reuse, and recycle." They sorted their trash, bought reusable bags, and trusted that individual actions could help fix a global crisis. But what many don't realize is that this sense of responsibility was never evenly distributed. The system was deliberately designed to make the public feel accountable for pollution, while the real power lies with the corporations that continue to produce waste on an industrial scale. Consumers are encouraged to feel guilty for using a plastic straw, while the companies producing billions of them face little to no consequences. In this way, good intentions are manipulated into a smokescreen — one that obscures the urgent need to hold corporations truly responsible.

As long as recycling remains the go-to symbol of sustainability, more ambitious and effective efforts risk being overlooked. But not all hope is lost. Small towns

PHOTO VIA ADOBE STOCK

across the country have been pioneering grassroots refill programs that encourage and provide people with resources to refill things like food containers, cleaning supply bottles, and personal care items like shampoo and conditioner. Large-scale bans now exist on single-use plastics in major cities, and some grocery stores are piloting package-free shopping aisles, while the scientific community is working on biodegradable alternatives to traditional packaging.

Notably, there is growing momentum around Extended Producer Responsibility (EPR) laws — policies that require manufacturers to take full accountability for the lifecycle of their products, including disposal and recycling. Unlike voluntary recycling initiatives, EPR makes corporations legally and financially responsible for the waste they produce, giving them a powerful incentive to reduce packaging and to design for reuse.

Here in Boston, Northeastern's Husky Environmental Action Team has worked to expand sustainable practices beyond recycling. They created a sub-section of their club dedicated to creating better composting practices throughout Northeastern, and they achieved it! There are now composting bins all over campus. These changes don't just shift behavior; they shift where the power lies. Real progress is happening, just not always in the spotlight.

This is by no means a call to abandon recycling; rather, it's a call to see it for what it is: a small piece of a much larger puzzle. Recycling can still play a role, but it should no longer be the centerpiece of our environmental efforts. By spreading awareness about its limitations, we can begin to break the cycle of guilt and inaction that's been unfairly placed on individuals. Real change means demanding more from the corporations that create the problem, supporting policies that hold them accountable, and amplifying the solutions that already exist but don't get enough attention. If we can shift the narrative — and with it, the power — then a more sustainable future isn't just possible, it's within reach. Hope doesn't lie in one blue bin; it lies in all of us pushing for something better.

CONGRATS

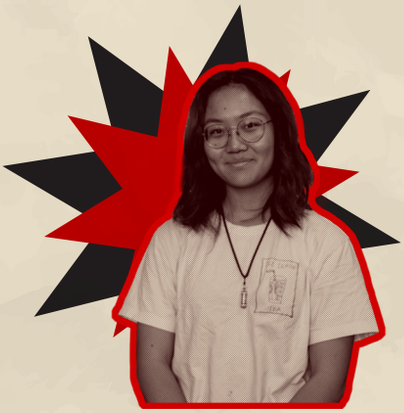
NU Sci *SENIORS*

GABRIELLE WEINER

Gabrielle is so thankful to have had three wonderful years with NU Sci — first as a writer, then treasurer, and finally as president of the club. She plans to work as a medical assistant for the next year while she completes her PA school applications. Gabrielle appreciates all of her writers, designers, and fellow e-board members and feels bittersweet as their time together comes to an end. She sends her love and best wishes to the club!

GIULIA WALKER

Giulia has been a part of the magazine since her sophomore year, beginning as a designer and working her way up to Senior Designer and ultimately Head. Through this experience, she developed a strong passion for editorial work, which led her to a role as an Editorial Designer for The Boston Globe. She has loved being part of the editorial process and creating pages for articles that are thoughtfully drafted, researched, and edited. Giulia is excited to graduate and move on to the next chapter.



JIAJIA FU

From chasing migrating sandhill cranes to designing synthetic proteins, JiaJia's love of nature and science has grounded her life. Since freshman year, NU Sci gave her a space to grow as a science communicator and explore beyond her comfort zone. NU Sci also found a job for her beloved camera whose lens through which she sees the world. As Head of Photography, she found joy in building a team, capturing seaside adventures, and bringing creative work to life. NU Sci gave her not just a voice, but a community united by curiosity. JiaJia is excited for her next adventure out west as a Molecular Engineering PhD student at the University of Washington.

ETHAN SZETO

In the spring of 2023, Ethan joined the NU Sci Webteam, working on his dreams of growing the team, building a new site, and providing a meaningful learning experience to the members. In 2026, he accomplished one of his goals by launching a completely new website, with the help of a large and devoted software team. Not everything he set out to do might have been accomplished on paper, but he's more than grateful for the family he found and the memories he made at NU Sci. Special thanks to Raisa Bhuiyan for always believing in me.

SASHA VOLKOVA

Sasha Volkova is a fourth-year student at Northeastern University studying Behavioral Neuroscience with a minor in Biochemistry. She currently serves as Treasurer of NU Sci and has been involved with the magagzine since her sophomore year. Sasha is particularly interested in the intersection of neuroscience, metabolism, and aging, and after graduation hopes to pursue further education focused on gut–brain interactions and longevity science. Outside of NU Sci and her coursework, she enjoys practicing yoga, creating and appreciating art, and cooking with friends.

EMILY XU

Emily has been a part of NU Sci as a writer since her sophomore year, started editing her junior year, and then became a Senior Editor as well. NU Sci has shaped her undergraduate experience by allowing her to explore scientific innovation and communication. Outside of school, Emily enjoys working with children, traveling, reading, and sewing. Following graduation, Emily will be applying to dental school in the hopes of becoming a dentist who makes oral healthcare more accessible to underserved communities!

CAROLINE GABLE

Drawn by her love of creative writing and science, Caroline joined NU Sci in her first semester at Northeastern. She went on to serve as both a journalist and senior editor, covering topics from galaxy clusters and llama-led glacial restoration to stress and giant squid mythology. She's deeply grateful to NU Sci for providing her a space to be curious about topics she would otherwise rarely learn about in her health science and psychology coursework. Caroline encourages readers and writers alike to stay curious and passionate, ask questions, and respect each other and our Earth. After graduation, she hopes to go backpacking, spend time with the people she loves, and finally finish the fantasy novel she started in fifth grade!

SASHI NALLAPATI

Sashi is a fourth-year Chemistry student and began writing for NU Sci at the start of her sophomore year at Northeastern. As a dedicated student and avid consumer of science writing, she loves helping to tweak writers' articles and finding their unique and individual voices in their writing. Sashi hopes to encourage scientific curiosity and communication through her involvement in NU Sci. Outside of school, you can find her at a cafe, reading, traveling, or crafting.

MACKENZIE HEIDKAMP

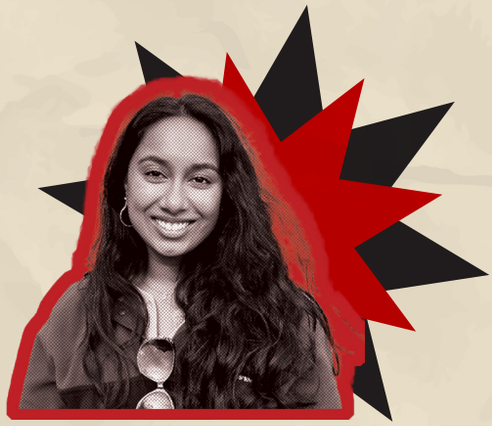
Eager to find a creative outlet outside of her science coursework, Mackenzie joined the magazine during her first semester of freshman year as a writer. While her academic interests center on chemistry, biology, and medicine, NU Sci gave her the opportunity to explore a much wider range of scientific topics. From quantum mechanics to the Great Emu War, writing for the magazine has allowed her to dive into material she never expected to encounter. As an editor, Mackenzie loved to help others in using their voices to make complex science more accessible.

MIKAYLA TSAI

Mikayla, a graduating Behavioral Neuroscience major, joined NU Sci her sophomore year. She contributed as a writer and editor for the Sunday Sci Newsletter. At the start of senior year, she became an editor, and has thoroughly enjoyed the process of learning and exploring various topics that pique her interest. Outside of NU Sci, Mikayla is a pre-med student who has conducted neurobiology research focusing on early-life adversity and cue learning. After graduation, she plans to work in a clinical allergy and asthma practice before applying to medical school.

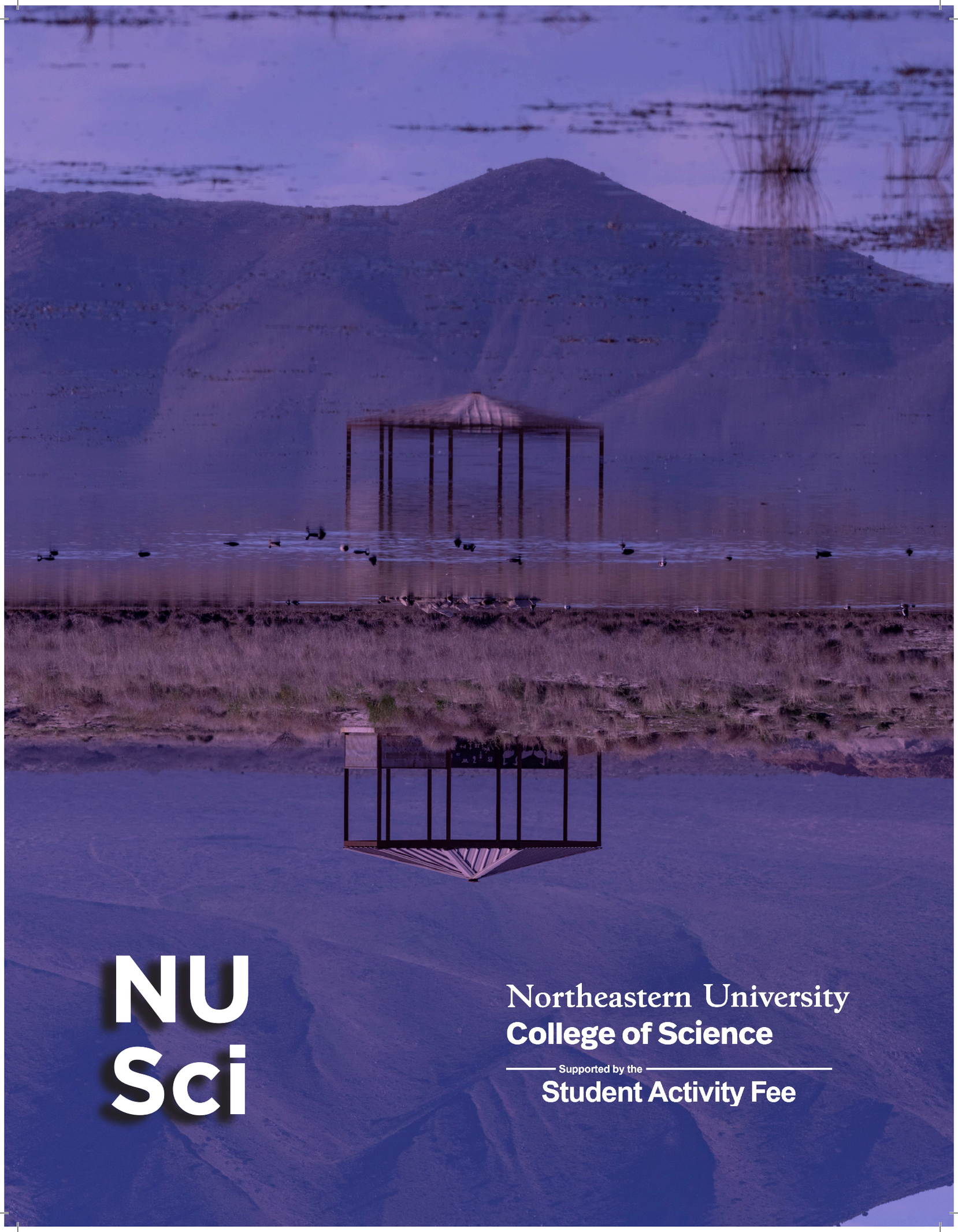
SAI TUMMALA

A lover of science and storytelling, Sai joined NU Sci wanting to make scientific research accessible, engaging, and human. For her, NU Sci became more than a magazine — it became a space where science could be thoughtful, inclusive, and reflective of diverse lived experiences. Outside of NU Sci, Sai is a cell and molecular biology major with a minor in psychology on the pre-med track. She hopes to pursue a career in medicine where she can advocate for underserved communities and continue bridging gaps between science and society. She would like to thank the NU Sci team for showing that science is most powerful when everyone feels invited into the conversation.



MAGGIE EID

Through NU Sci, Maggie treasured the opportunity to pursue her passion for increasing science accessibility. Joining the web team in her first semester, Maggie became an editor in her third year, eager to get more involved with the magazine's writing side. She was incredibly grateful for NU Sci's collaborative environment and the opportunities to support and learn from her peers. Maggie loved researching topics within her major, environmental science, as well as exploring other fields. Believing that science communication is an essential skill, she cherishes the writing experience she gained as an editor and author, sharing the wonder and importance of science. After graduating in December, Maggie is assisting with research at the University of New Hampshire and working as a microbiologist at a lab testing company.



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